

多环境下的LLM Agents 应用与增强

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01 大模型与智能体

02 多模具身智能体

03 推理密集智能体

04 科学领域智能体

01

LLM and Agents

它的定义，框架，与挑战

■ 什么是Agent?



Agent的定义



“An **agent** is anything that can be viewed as **perceiving** its **environment** through **sensors** and **acting** upon that environment through **actuators**.
—— Stuart J. Russell and Peter Norvig”

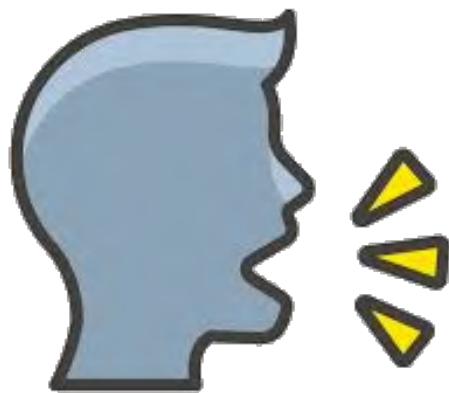
■ 简而言之。。。



“An *agent* is a system
that can help *complete*
tasks intelligently.”

■ Just Do It

But how?



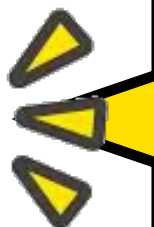
We need tools



■ Just Say It

But how?

We need tools

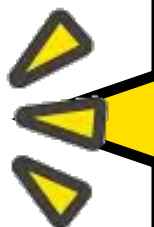


■ 幕后主脑

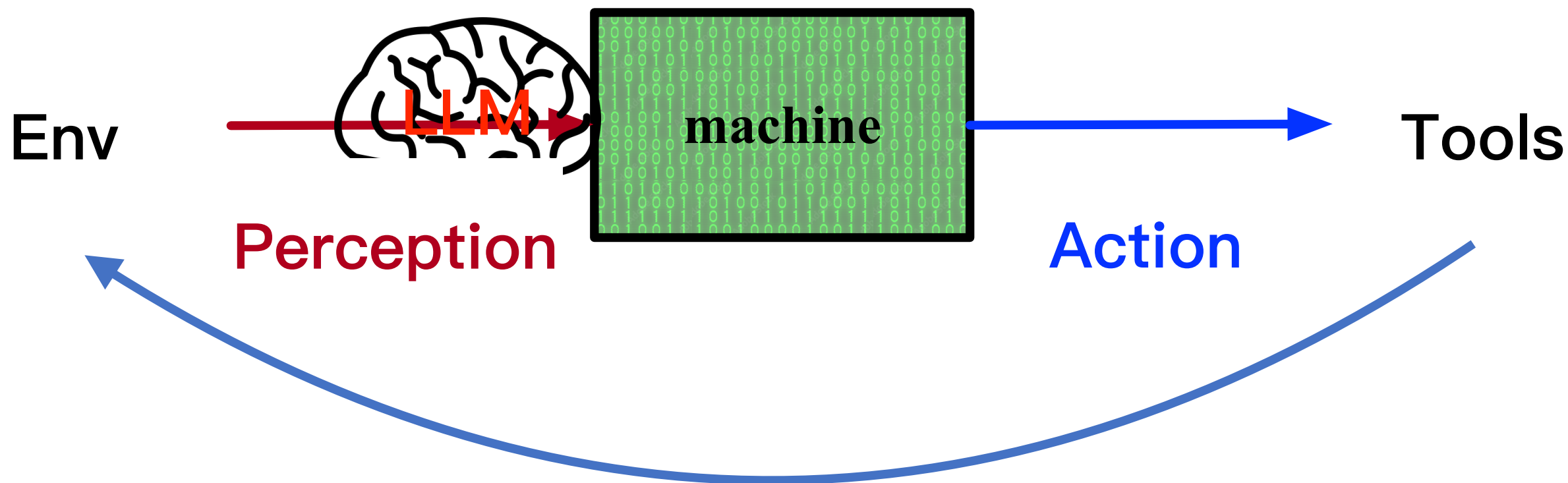
But how?

We need LLM

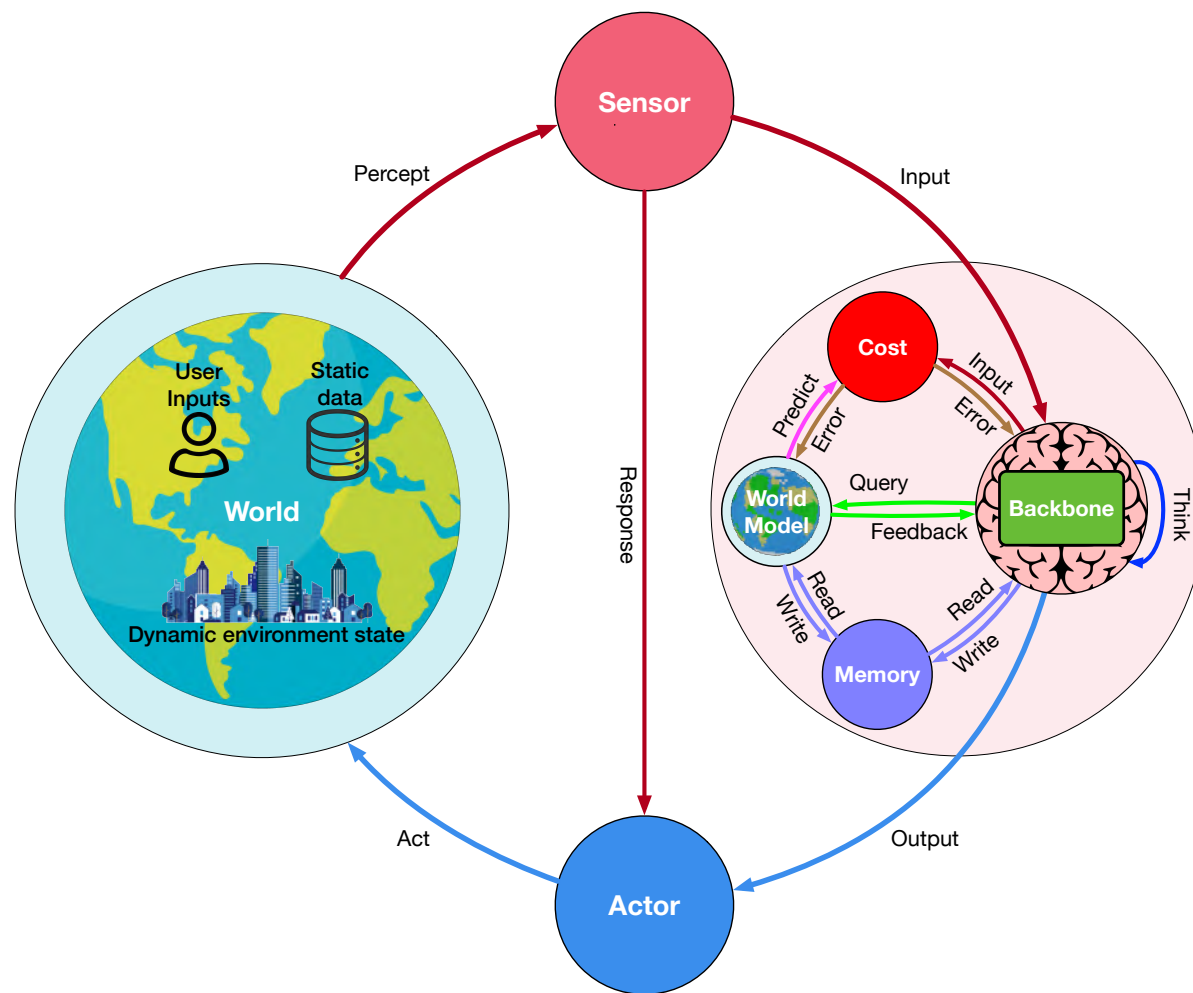
We need tools



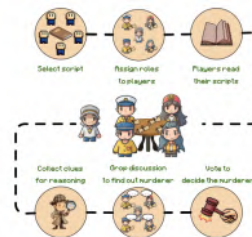
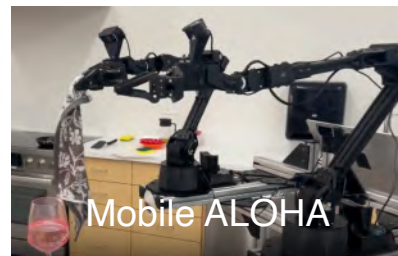
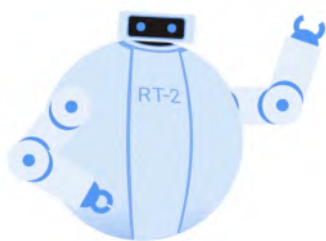
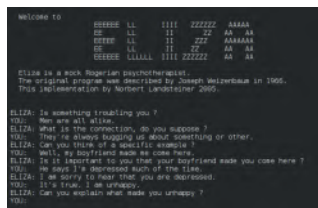
■ 基于LLM的自然语言处理



Agent系统框架



一些现存的 Agents



ThinkThrice



Mobile-Agent

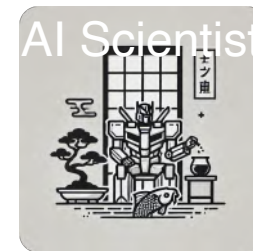
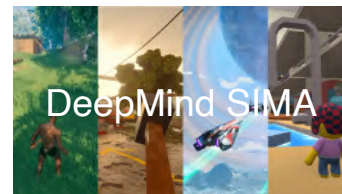
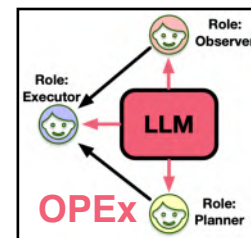
● Open Interpreter



HoneyBee

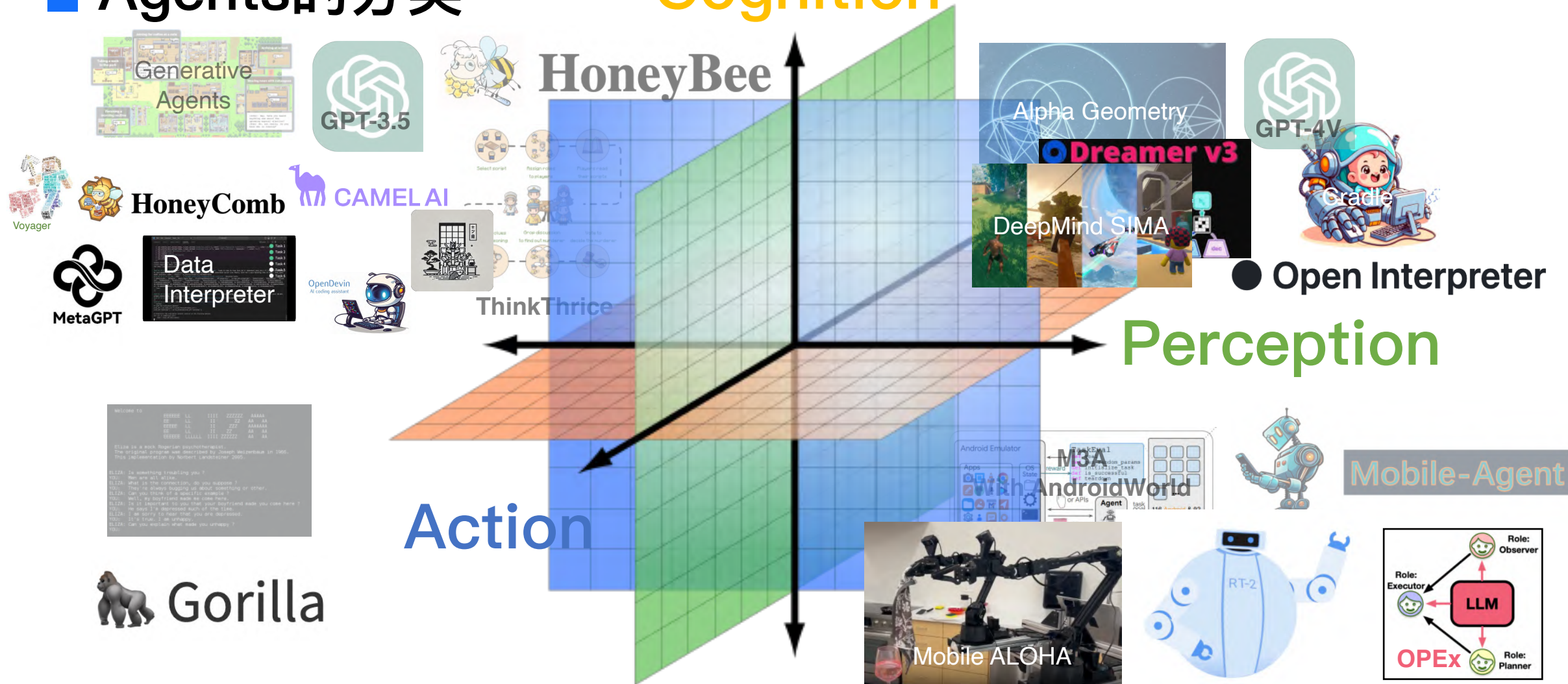


HoneyComb

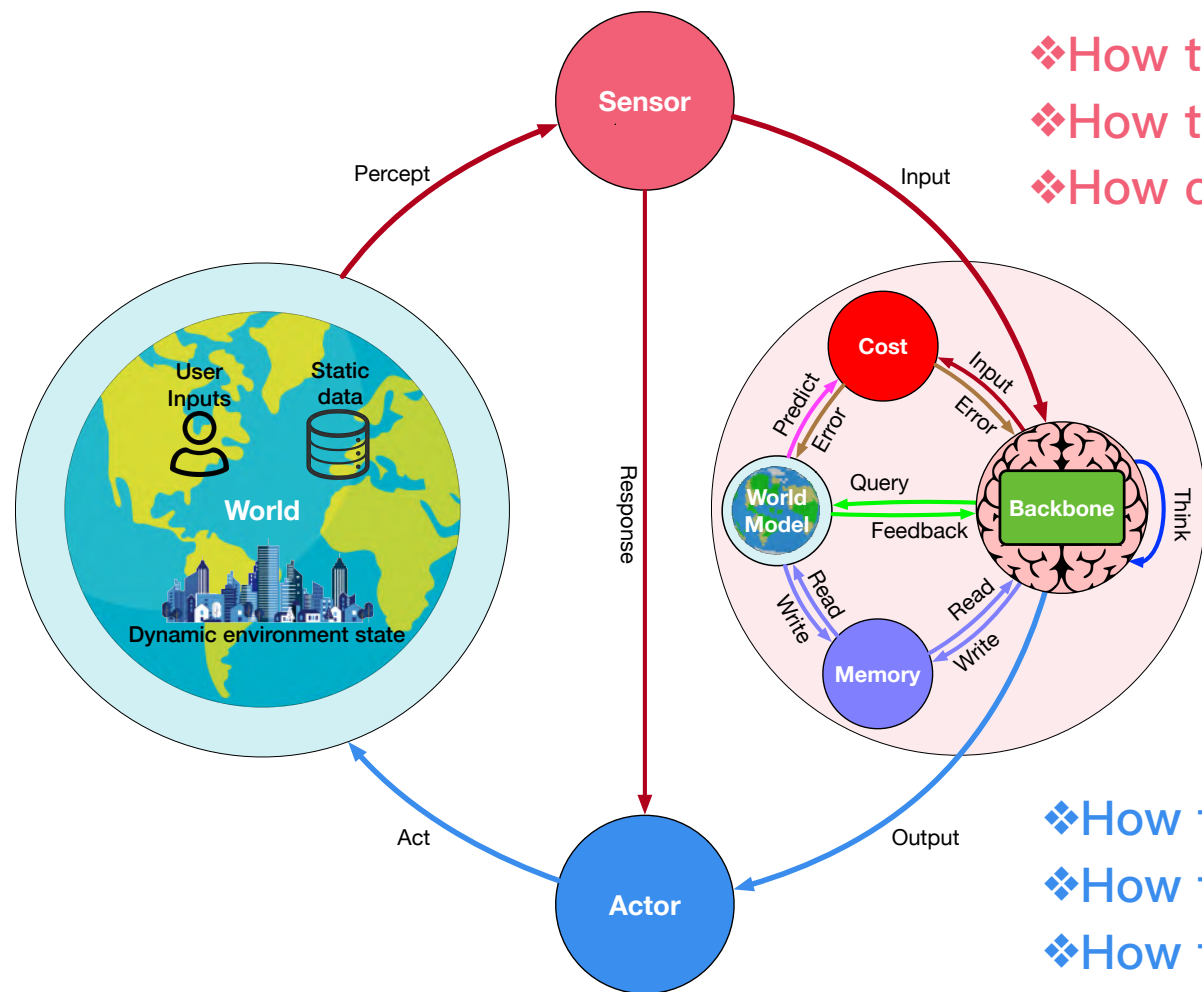


Agents的分类

Cognition



LLM Agent的一些核心挑战



- ❖ How to represent and align multimodal input signals?
- ❖ How to achieve real-time perception in dynamic settings?
- ❖ How can agents handle incomplete or noisy data robustly?

- ❖ How to perform complex tasks?
- ❖ How to deal with unseen tasks?
- ❖ How to learn and utilize domain knowledge?

- ❖ How to effectively execute actions?
- ❖ How to search huge action space?
- ❖ How to design and evolve tools?

研究轨迹

“任何事物是与一切其他事物关系的总和”

- 语言塑造思维
 - 如何挖掘和利用不同类型和层次的文本关系来提高语言理解与生成?
 - 代表研究: DLG4NLP, QG4QA
- 自然语言处理

- 知中有行, 行中有知
 - 如何构建世界模型, 提升Agent System的各个模块(感知, 认知, 行为)?
- 具身学习

多模态

- 跳出中文屋
- 如何表征多模态信息及其之间的关系, 如何对齐和融合不同模态?

- AI赋能科学发现
- 构建科学智能体, 发掘因果关系, 加速科学发现
- 代表研究: AI4MatSci

AI for Science

AI Agent & Society

- 三个臭皮匠, 顶个诸葛亮
- 构建Agent & Multi-Agent System解决复杂任务

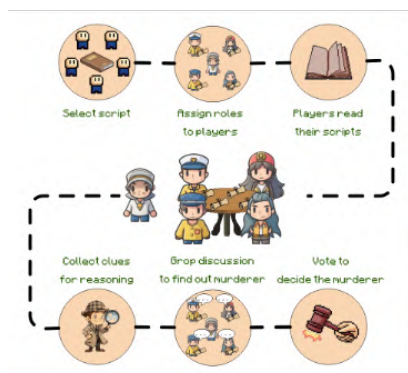
研究轨迹

“人是一切社会关系的总和”

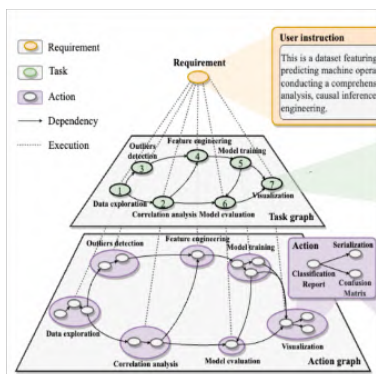


近期成果

LLM and Agents

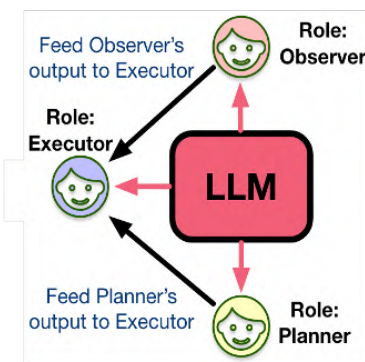


Jubensha
Improve
multi-agent
reasoning



**Data
Interpreter**
LLM Agent
For Data
Science

多模态与具身学习



OPEx
Analysis LLM
Agents for
embodied
learning



VCR
Visual
Caption
Restoration

近期成果

AI for Materials Science



HoneyBee
LLM for
materials
science



HoneyComb
LLM Agent for
materials
science

02

Multimodal Embodied Agents

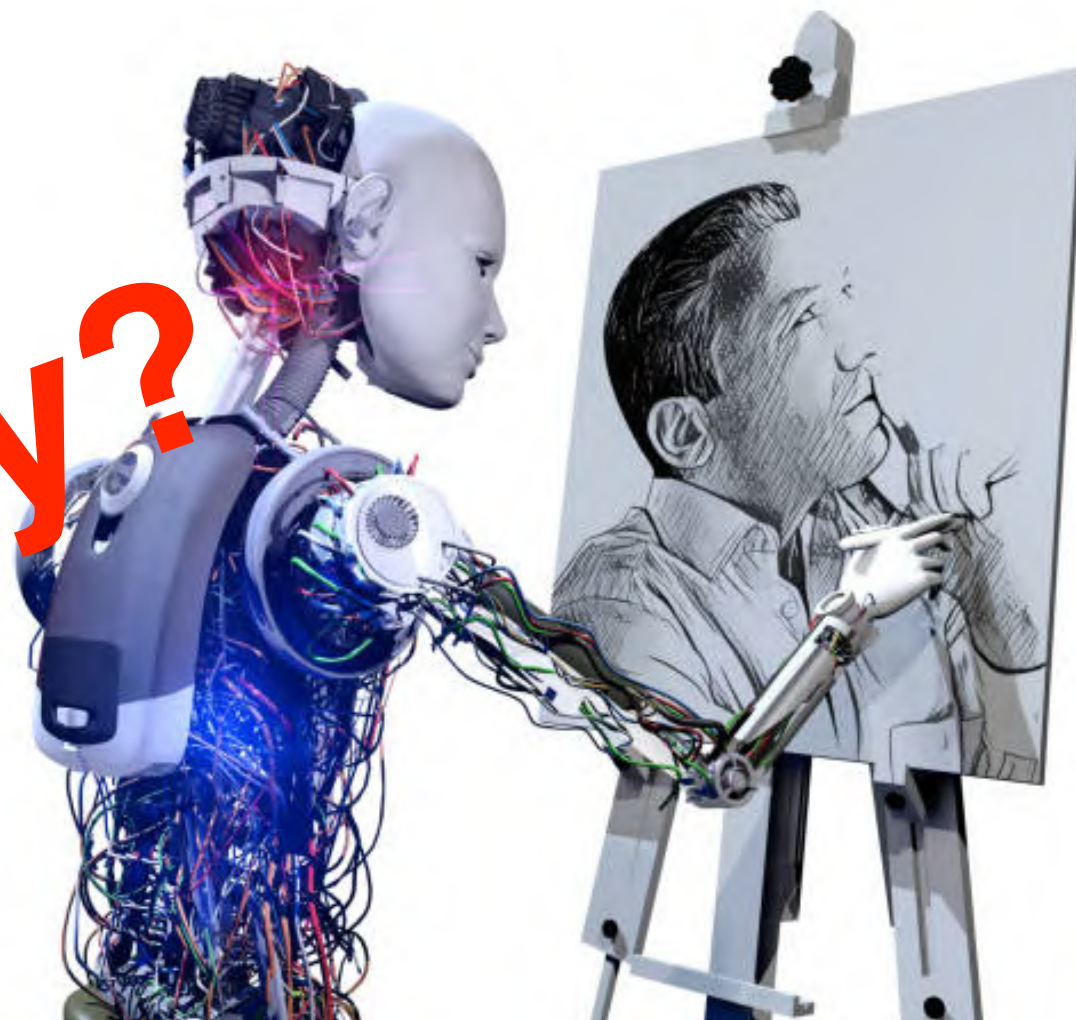
我们离具身智能助手还有多远？



What We Want v.s. What We Get



Why?



■ Case Study: Embodied Instruction Following (EIF)

★ Current Scene: FloorPlan1



🔍 Filter Scenes

- ☒ iTHOR
 - ☒ Kitchens
 - ☒ Living Rooms
 - ☒ Bedrooms
 - ☒ Bathrooms
- ☒ RoboTHOR
 - ☒ Training
 - ☒ Validation



Case Study: Embodied Instruction Following (EIF)

EIF task types in ALFRED

Pick & Place	Pick Two & Place	Stack & Place	Examine in Light
Heat & Place	Clean & Place	Cool & Place	

One task example

Task arguments
Item(s): Ladle Receptacle: Table Scene #: 27

One expert annotation
High-level instruction: Place a clean ladle on a table

Low-level instruction: Turn around and veer left to the white table.

Ego-centric RGB input:
t = 28

Low-level instruction: Pick up the ladle on the table, behind the tomato.

Ego-centric RGB input:
t = 61

Low-level instruction: Turn right and walk to the sink

Ego-centric RGB input:
t = 74

Low-level instruction: Wash the ladle in the right sink basin.

Ego-centric RGB input:
t = 83

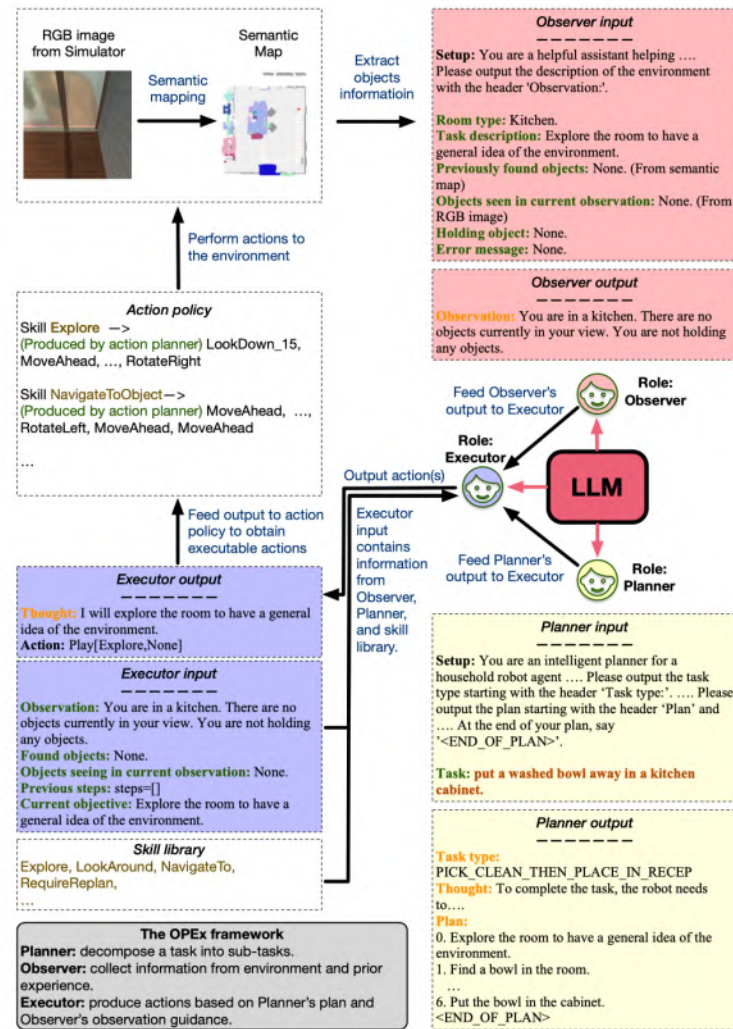
Low-level instruction: Turn left and walk back to the white table.

Ego-centric RGB input:
t = 117

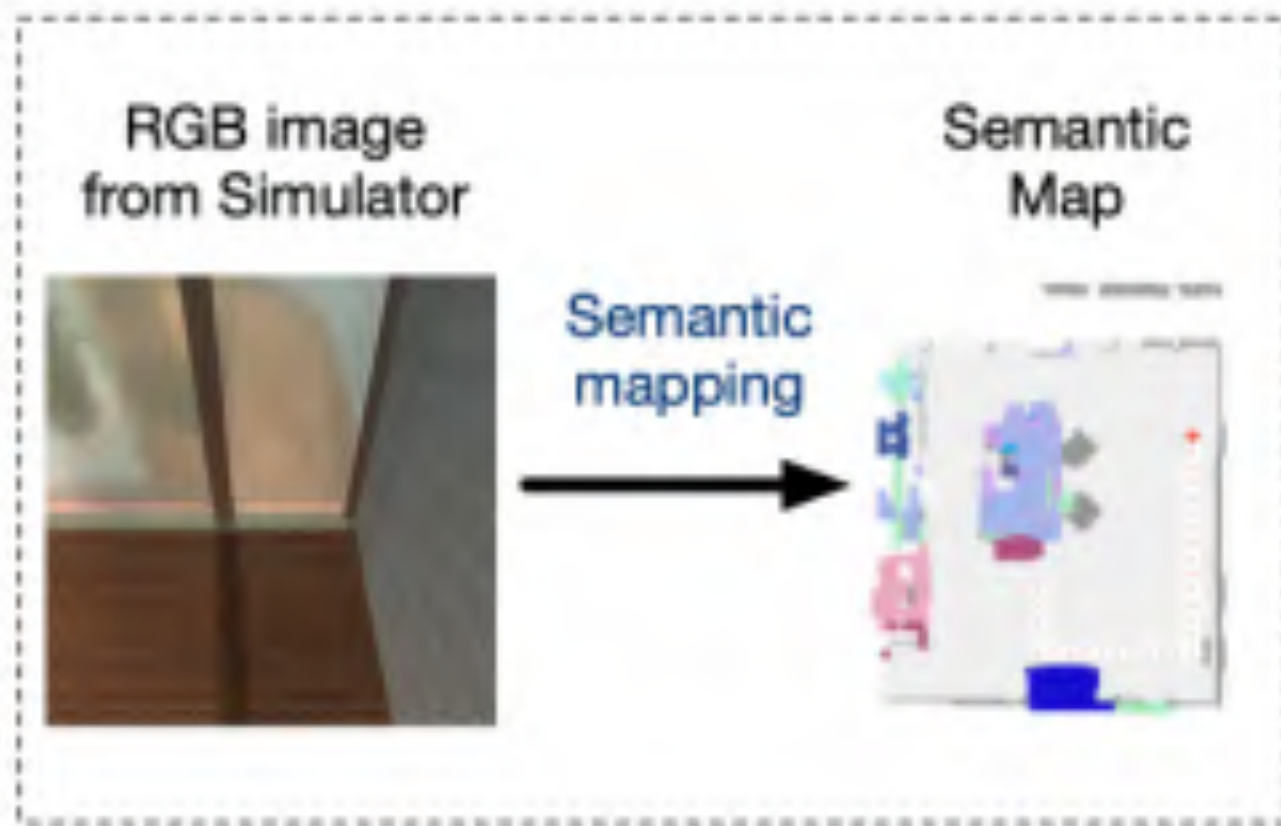
Low-level instruction: Place the ladle on the table in front of the white mug.

Ego-centric RGB input:
t = 164

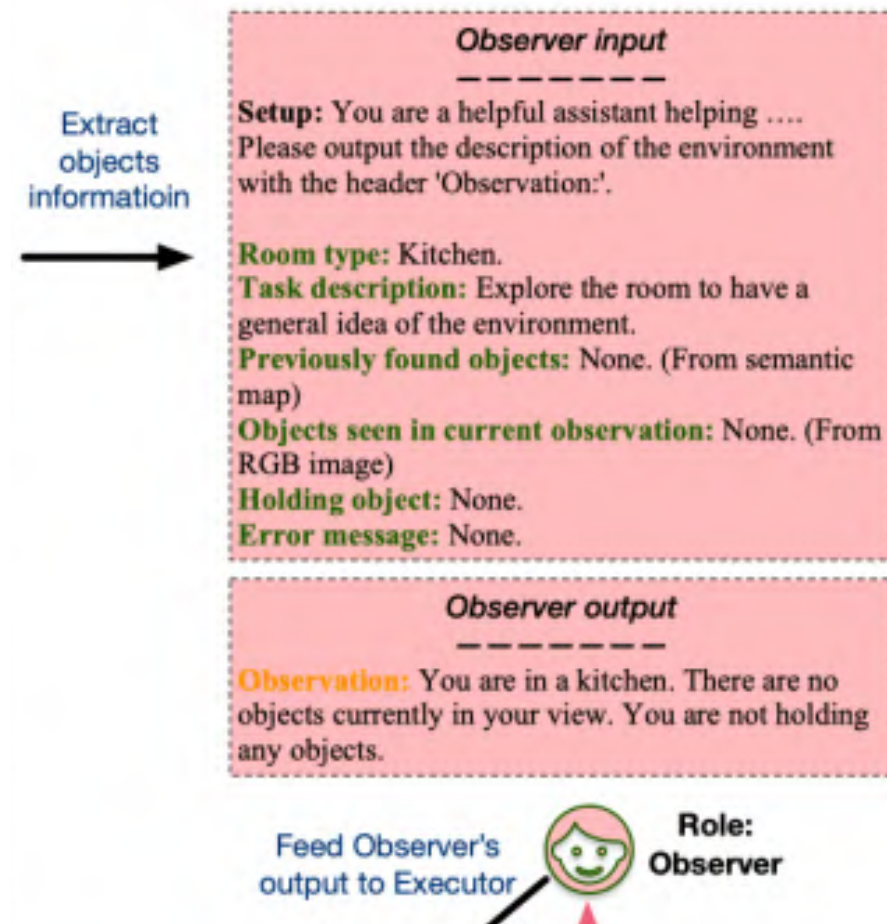
OPEx: A Component-Wise Analysis



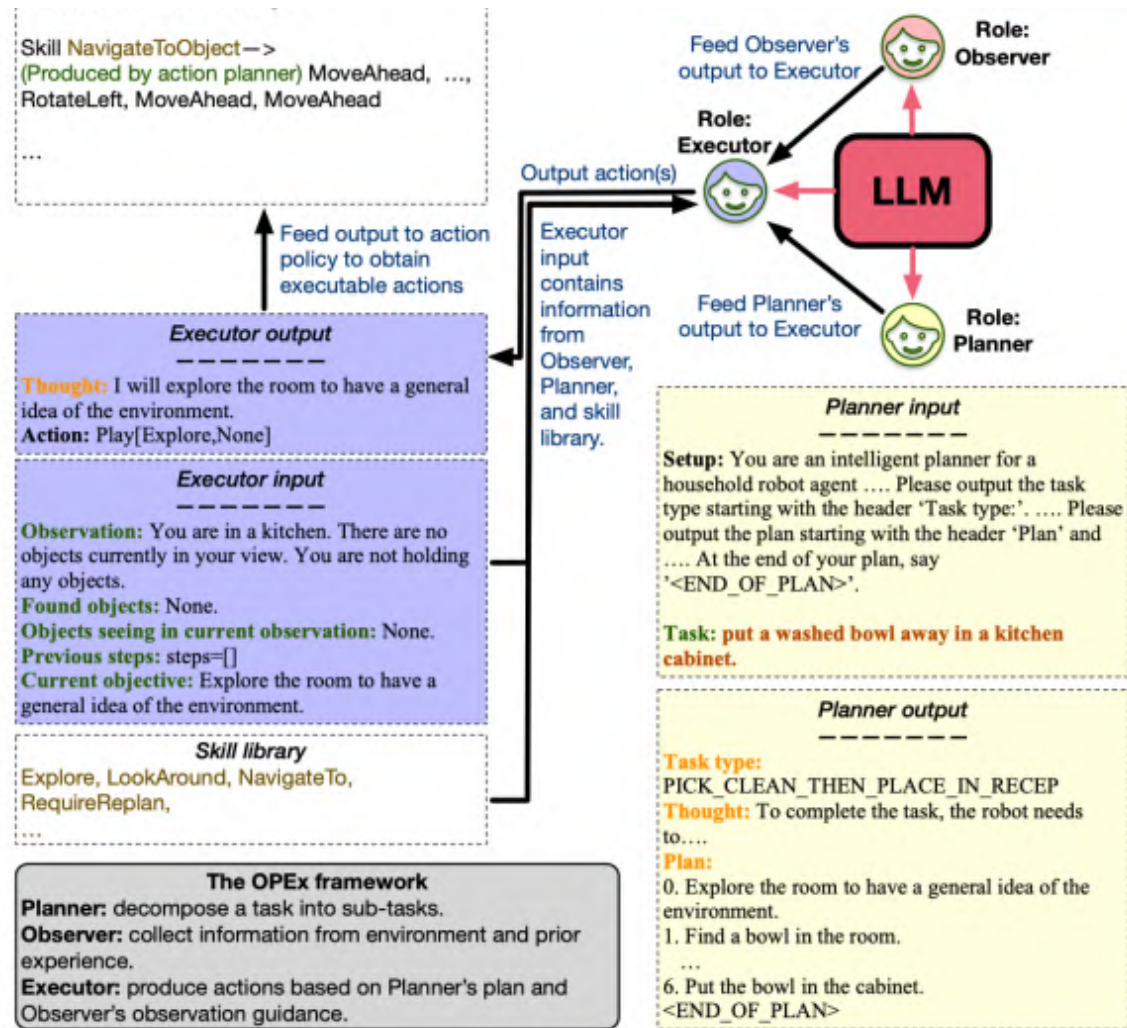
■ OPEx: A Component-Wise Analysis



■ OPEx: A Component-Wise Analysis

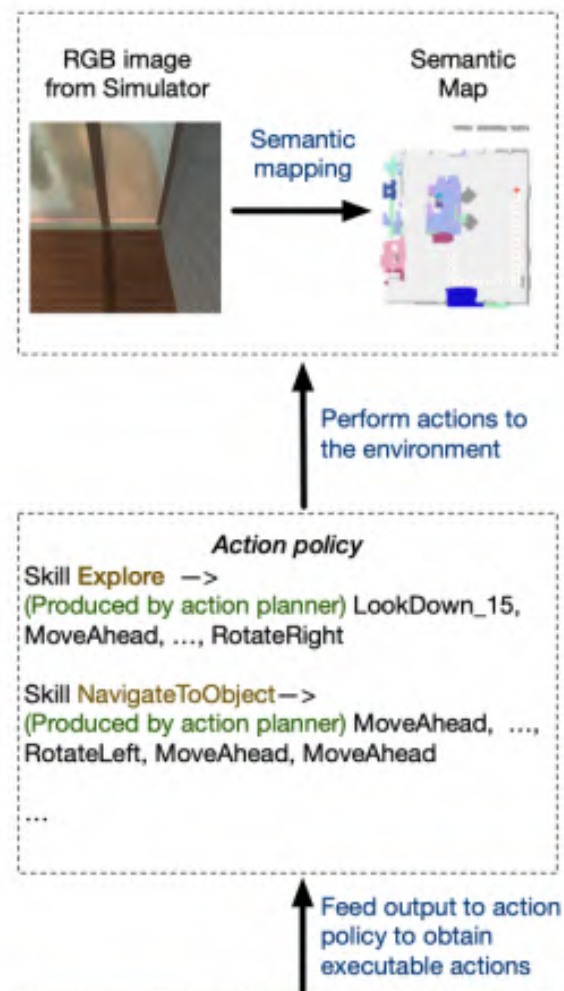


OPEx: A Component-Wise Analysis



Reference: Shi et al. "OPEx: A Component-Wise Analysis of LLM-Centric Agents in Embodied Instruction Following", ACL 2024.

■ OPEx: A Component-Wise Analysis



Insights from OPEx

Component-wise Analysis

Method	Valid Uneen			
	PLWGC	GC	PLWSR	SR
Influence of perception models				
OPEx	13.48	48.61	9.08	35.91
OPEx-S	16.52	51.28	11.38	40.80
OPEX-P	23.72	66.17	17.43	59.43
Influence of action policies				
OPEx-S	16.52	51.28	11.38	40.80
-semantic map M'_t	12.37	45.41	8.06	36.17
-slice replay	12.64	45.25	8.35	37.39
-traversable goal	11.77	43.49	7.09	34.50
Influence of LLM-based modules				
OPEx-S	16.52	51.28	11.38	40.80
-Planner	8.10	40.16	5.72	30.57
-Observer	13.41	45.62	8.58	37.76
Influence of introduced skills				
OPEx-S	16.52	51.28	11.38	40.80
-LookAround	12.56	45.37	8.26	37.76
-Explore	13.29	47.32	8.44	38.25
Influence of prior knowledge (On ALFWorld)				
ReAct	-	-	-	66
OPEx	-	-	-	73
OPEx-L	-	-	-	78
OPEx-H	-	-	-	84

As the bases for further reasoning, planning, and acting, a **precise representation of the Env** is critical.

Heuristic action policy is a **vanilla solution**.
- Any better learning-based method?

Significant potential for better grounding.

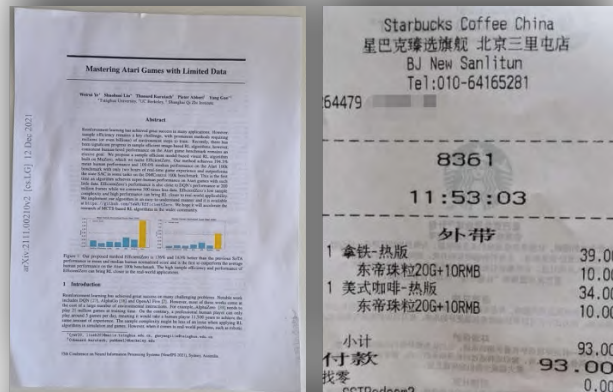
Reducing planning horizon is vital.

The **knowledge gap** between **pre-trained LLMs** and the **target domain** should be bridged.

Low Demand for In-Domain Data

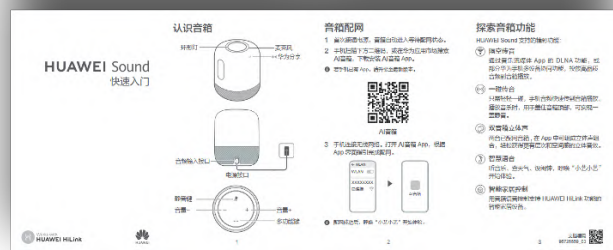
Method	SR	GC	PLWSR	PLWGC
OPEx	38.12	46.13	9.03	13.45
FILM	0.00	12.18	0.00	2.78

We Need to Strengthen Multimodal Perception



文档理解 (Doc Understanding)

❖目前大多数文档 LLM评测中，答案仅来自于图中的文字部分



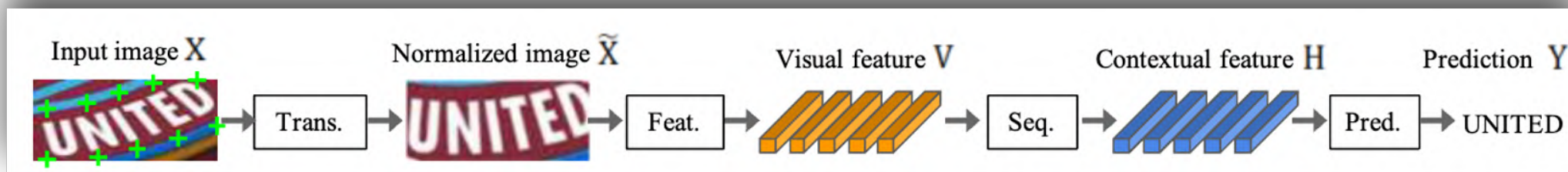
场景文字识别 (Scene Text Recognition)

❖可能直接通过OCR识别文字图象，不需要自然语言知识



视觉问答 (VQA)

❖可能不需要图像信息即可回答



■ 视觉字幕恢复 (VCR)



至大论，公元1400年编纂的一部数学、天文学专著，提出了行星的运动路径。直到中世纪和文艺复兴早期，该书被伊斯兰和欧

图像中被覆盖的文本是什么？请在不输出解释的情况下还原被覆盖的文本。

❖答案取决于图像+文字图像+像素级文字提示+自然语言问题

❖优势：需要进行图像/文字图像/自然语言三者的对齐

■ 视觉字幕恢复 (VCR)

图像中被覆盖的文本是什么？请在不输出解释的情况下还原被覆盖的文本。 ———→ 自然语言问题



图像

全大论，是哥白尼在1543年编纂的一部数学、天文学专著，提出了日心说，认为行星的运动路径。直到中世纪和文艺复兴早期，该书一直被伊斯兰和欧

文字图像

像素级文字提示

■ 人类在视觉补全上的复杂机制

Dynamic representation of partially occluded objects in primate prefrontal and visual cortex

Amber M Fyall^{1†}, Yasmine El-Shamayleh^{2†}, Hannah Choi³, Eric Shea-Brown⁴, Anitha Pasupathy^{5*}

¹Department of Biological Structure, Washington National Primate Research Center, University of Washington, Seattle, United States; ²Physiology and Biophysics, Washington National Primate Research Center, University of Washington, Seattle, United States; ³Applied Mathematics, University of Washington Institute for Neuroengineering, University of Washington, Seattle, United States; ⁴Department of Applied Mathematics, University of Washington, Seattle, United States; ⁵Department of Biological Structure, University of Washington, Seattle, United States

Abstract Successful recognition of partially occluded objects is presumed to involve dynamic interactions between brain areas responsible for vision and cognition, but neurophysiological evidence for the involvement of feedback signals is lacking. Here, we demonstrate that neurons in the ventrolateral prefrontal cortex (vLPFC) of monkeys performing a shape discrimination task respond more strongly to occluded than unoccluded stimuli. In contrast, neurons in visual area V4 respond more strongly to unoccluded stimuli. Analyses of V4 response dynamics reveal that many neurons exhibit two transient response peaks, the second of which emerges after vLPFC response onset and displays stronger selectivity for occluded shapes. We replicate these findings using a model of V4/vLPFC interactions in which occlusion-sensitive vLPFC neurons feed back to shape-selective V4 neurons, thereby enhancing V4 responses and selectivity to occluded shapes. These results reveal how signals from frontal and visual cortex could interact to facilitate object recognition under occlusion.

DOI: <https://doi.org/10.7554/eLife.25784.001>

Abstract: Recognizing highly occluded objects is believed to arise from the interaction between the brain's vision and cognition-controlling areas, although supporting neuroimaging data are currently limited. To explore the neural mechanism during this activity, we conducted an occlusion object recognition experiment using functional magnetic resonance imaging (fMRI). During magnet resonance examinations, 66 subjects engaged in object recognition tasks with three different occlusion degrees. Generalized linear model (GLM) analysis showed that the activation degree of the occipital lobe (inferior occipital gyrus, middle occipital gyrus, and occipital fusiform gyrus) and dorsal anterior cingulate cortex (dACC) was related to the occlusion degree of the objects. Multivariate pattern analysis (MVPA) further unearthed a considerable surge in classification precision when dACC activation was incorporated as a feature. This suggested the combined role of dACC and the occipital lobe in occluded object recognition tasks. Moreover, psychophysiological interaction (PPI) analysis disclosed that functional connectivity (FC) between the dACC and the occipital lobe was enhanced with increased occlusion, highlighting the necessity of FC between these two brain regions in effectively identifying exceedingly occluded objects. In conclusion, these findings contribute to understanding the neural mechanisms of highly occluded object recognition, augmenting our appreciation of how the brain manages incomplete visual data.

Keywords: fMRI; MVPA; PPI; dACC; occipital lobe; occluded object recognition

- ❖ 人类很擅长识别被部分遮挡的物体
- ❖ 人类在识别时会使用不同的脑区
- ❖ 在识别被遮挡的事物涉及大脑不同区域之间的复杂协同，这些区域与视觉和感知控制相关

VCR任务的独特性

❖设计初衷：解决视觉问答和场景文字识别中的问题

❖不能避免多模态识别：图像信息对于正确回答是必需的

❖不能通过OCR解决：被遮挡的文字仅保留像素级提示，无法通过OCR识别

❖具有唯一答案：与Masked LM不同，VCR中被遮挡的文字被露出的像素级提示唯一确定，可使用准确度（ACC）作为评测指标

❖设计理念：将图像中的文字作为第三种模态对待

❖图像中的文字具有和字符串类型自然语言/常规图像不同的特征，在多模态模型中应该单独加以考虑

❖设计实现：对于图像/文字来源及难度的高度灵活性

❖遮挡文字选择的灵活性：可选择遮挡特定token/句子/n-gram/POS Tag

❖遮挡区域选择的灵活性：可调节遮挡文字区域的空白框高度来控制任务难度

❖图像构建方式的灵活性：可选择在文字部分基础上有/无配套图像探究额外图像对模型影响

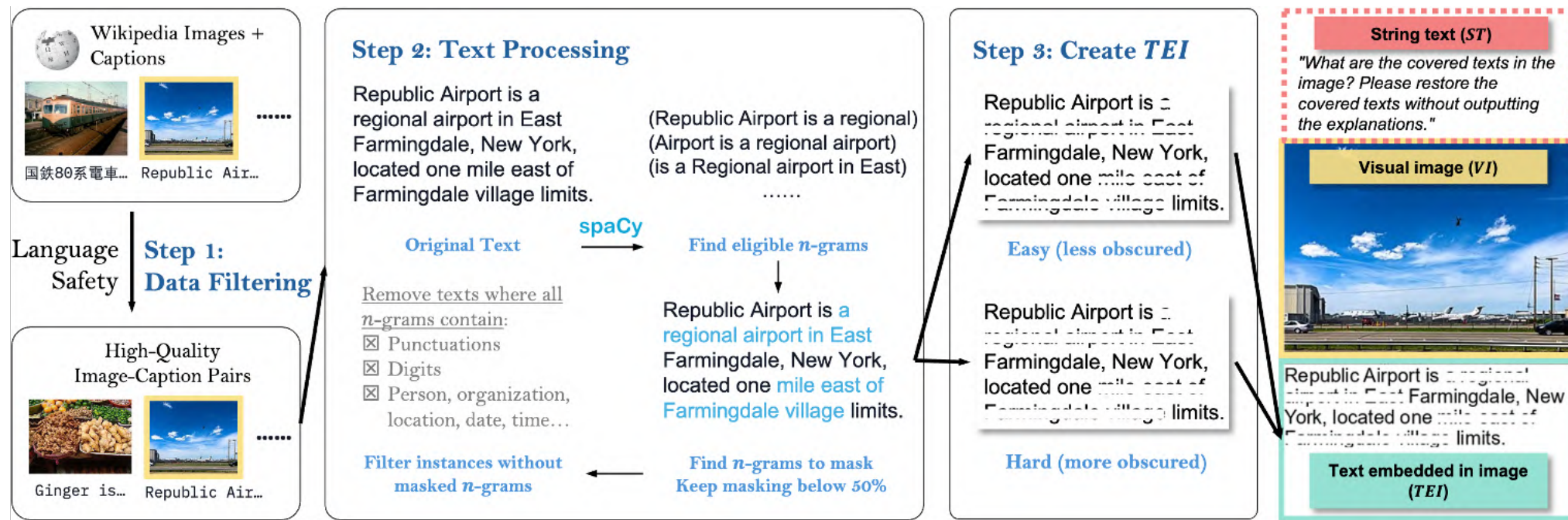
VCR-Wiki数据集构建

❖我们基于维基百科构建了适用于VCR任务的数据集**VCR-Wiki**

❖包含两种语言：简体中文和英文

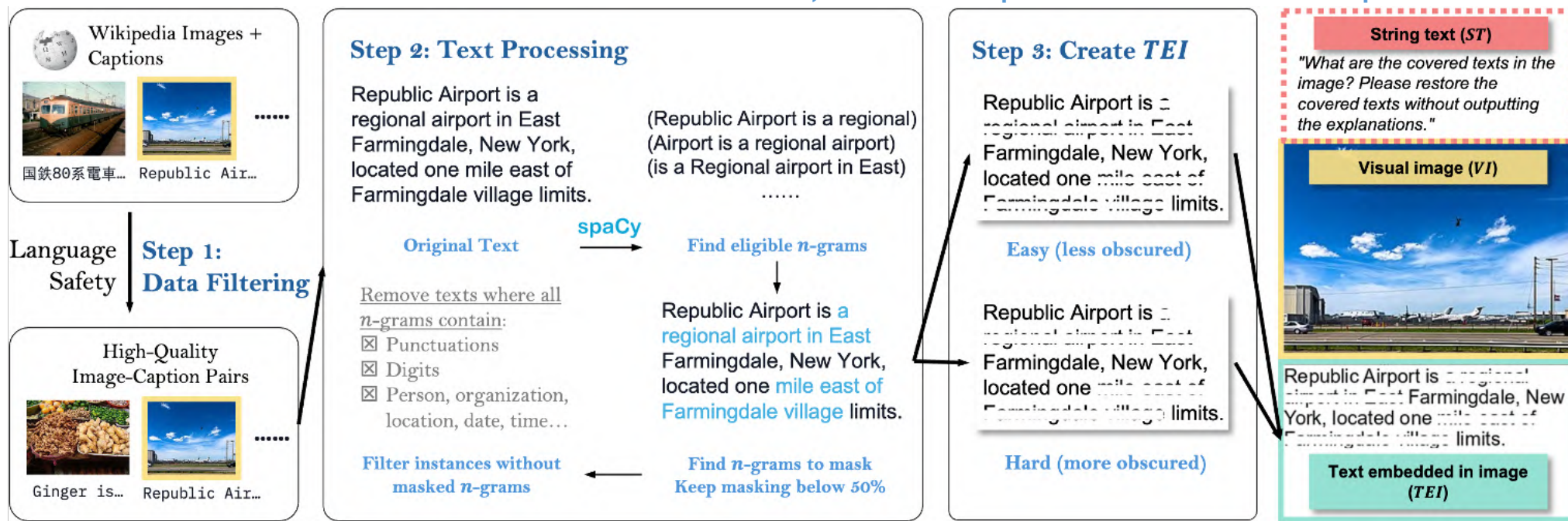
❖包含两种难度：简单（OCR无法完成的难度）和困难（遮挡文字上下仅各保留1-2个像素）

❖包含训练集+验证集+测试集，其中训练集可以作为多模态大模型SFT数据



VCR-Wiki数据集构建

1. **数据清洗**：基于wit-base数据集，去除不含zh/en的条目及部分敏感条目
2. **文本处理**：基于给定字体和字号，保留前5行维基百科介绍，通过spaCy选取其中不包含标点/数字/人名地名等的5-gram作为遮挡目标；筛去所有不包含任何遮挡目标的条目
3. **构建文字图像 (TEI)**：根据难度对遮挡目标进行不同程度遮挡
4. **拼接**：将视觉图像 (VI) 与文字图像 (TEI) 进行拼接，缩放至300px宽；筛去所有超过900px高的条目



英文VCR样例

简单



Lincoln is the luxury vehicle division of American automobile manufacturer Ford. Marketed among the top luxury vehicle brands in the United States, for

困难



Lincoln is the luxury vehicle division of American automobile manufacturer Ford. Marketed among the top luxury vehicle brands in the United States, for

中文VCR样例

简单



科西嘉是地中海西部的一座岛屿，也是法国的一个单领土集体，位于法国大陆部分的东南面，意大利半岛的西面，最近的地块是紧邻意大利的撒丁岛。科西嘉岛三分之二

困难



科西嘉是地中海西部的一座岛屿，也是法国的一个单领土集体，位于法国大陆部分的东南面，意大利半岛的西面，最近的地块是紧邻意大利的撒丁岛。科西嘉岛三分之二

实验观察

- ❖在VCR-Wiki上，对应语言的熟练使用者能够在简单和困难难度上取得超过90%的准确率，而**模型距离人类水平还有极大差距**
- ❖**目前开源模型整体弱于闭源模型**，但存在有开源模型（如CogVLM2）能够以19B参数量大幅超越相似大小的开源模型及部分闭源模型
- ❖VCR任务虽然看似简单，但其挑战涉及**模型分辨率压缩**（对像素级文本提示的保留），**模型推理**（利用上下文推断被覆盖文本）和**常识利用**（维基百科内容应已被各模型训练集覆盖，但效果仍然不好），在该评测上的提升还有很长的路要走
- ❖我们在持续更新新模型在VCR-Wiki上的效果，希望能**将VCR构建成为未来的视觉-语言模型（VLM）常用评测之一**

Resources about VCR



arXiv

[VCR Visual Caption Restoration](#)
[Arxiv-2406.06462](#)



GitHub

<https://github.com/tianyu-z/VCR>



Hugging Face

[VCR: Visual Caption Restoration](#)
[a vcr-org Collection \(huggingface.co\)](#)



机器之心报道

[Bengio团队提出多模态新基准，直指](#)
[Claude 3.5和GPT-4o弱点](#)



使用EvolvingLMMs-Lab/Imms-eval进行测试
[EvolvingLMMs-Lab/Imms-eval: Accelerating](#)
[the development of large multimodal models](#)
[\(LMMs\) with Imms-eval \(github.com\)](#)

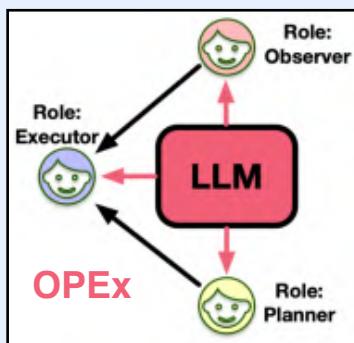


使用open-compass/VLM EvalKit进行测试
[open-compass/VLM EvalKit: Open-source](#)
[evaluation toolkit of large vision-language](#)
[models \(LVLMs\), support GPT-4v, Gemini,](#)
[QwenVLPlus, 50+ HF models, 20+](#)
[benchmarks \(github.com\)](#)

■ 小结：多模态与具身Agent

OPEx

目前 LLM-centered Embodied Agents 瓶颈在于多模态Perception的表征与具身Action的执行



VCR

VCR任务有助于提高多模态的表示学习，可成为视觉-语言模型（VLM）常用评测任务之一



03

Reasoning-Intensive Agents

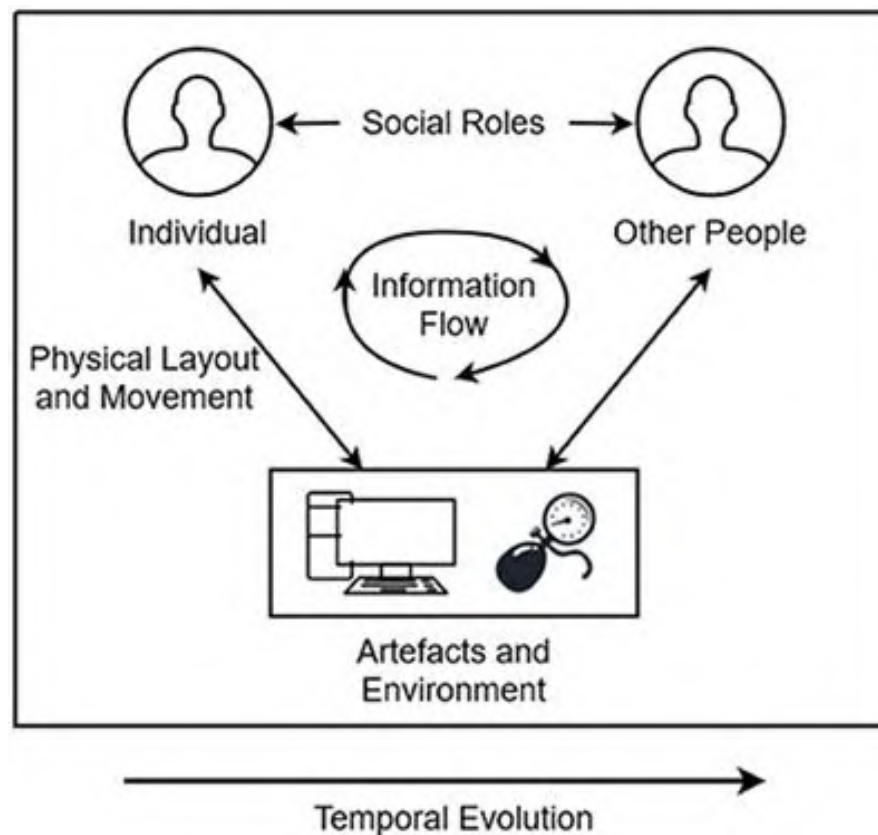
“体力”有待提高，“脑力”水平如何？



“人”是环境的一部分



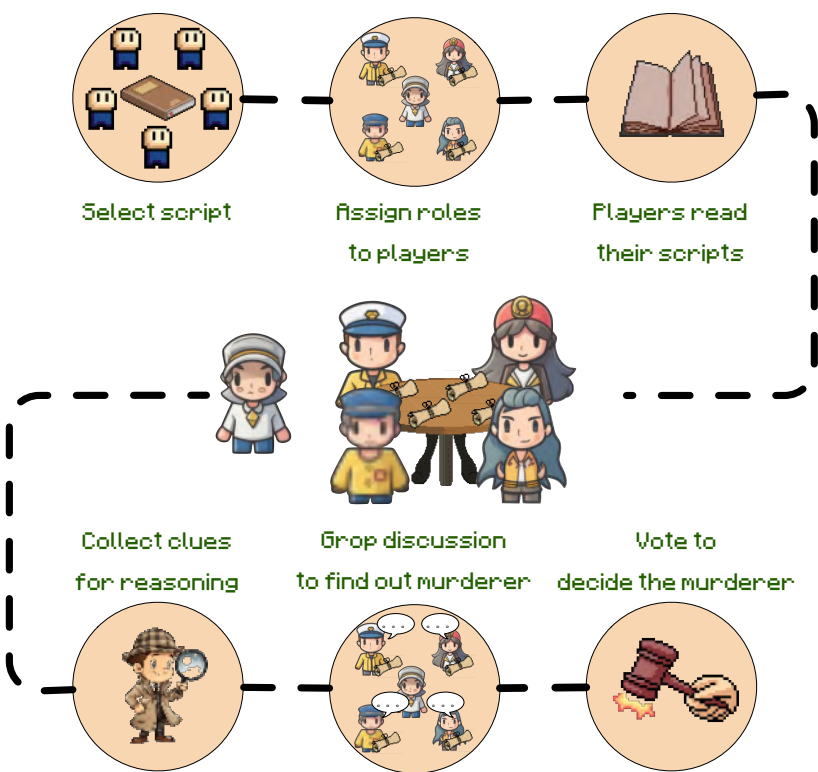
Pyramide à renverser - The poster shows a [social stratification](#) pyramid which symbolises [class society](#). At the top we can see King [Leopold II](#).



来骗！
来偷袭！

Diverse
Complexity!

■ 剧本杀 (Jubensha)



(a) Game flow of Jubensha detective game

多角色互动

复杂的推理和决策

语言理解和生成

多模态信息处理

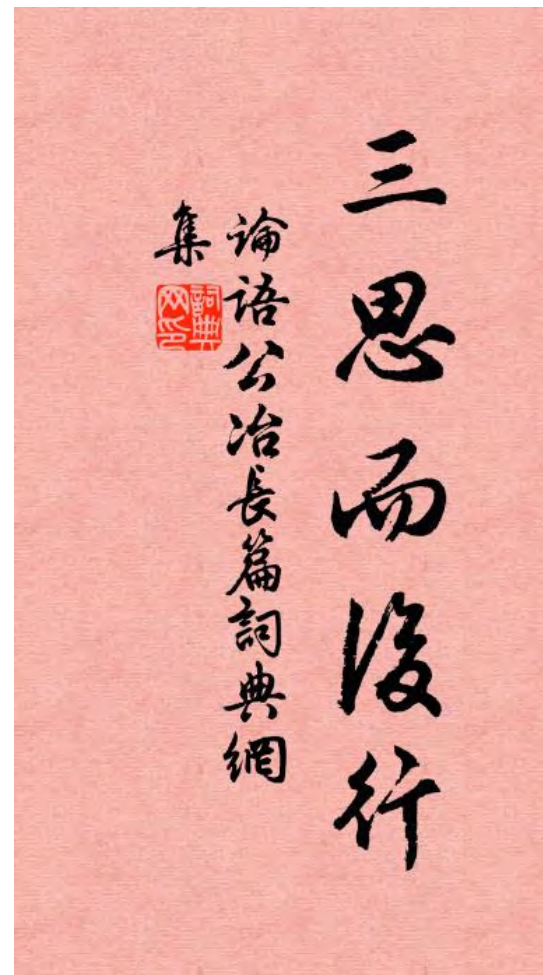
部分信息和不完全信息的处理

角色扮演和情境适应性

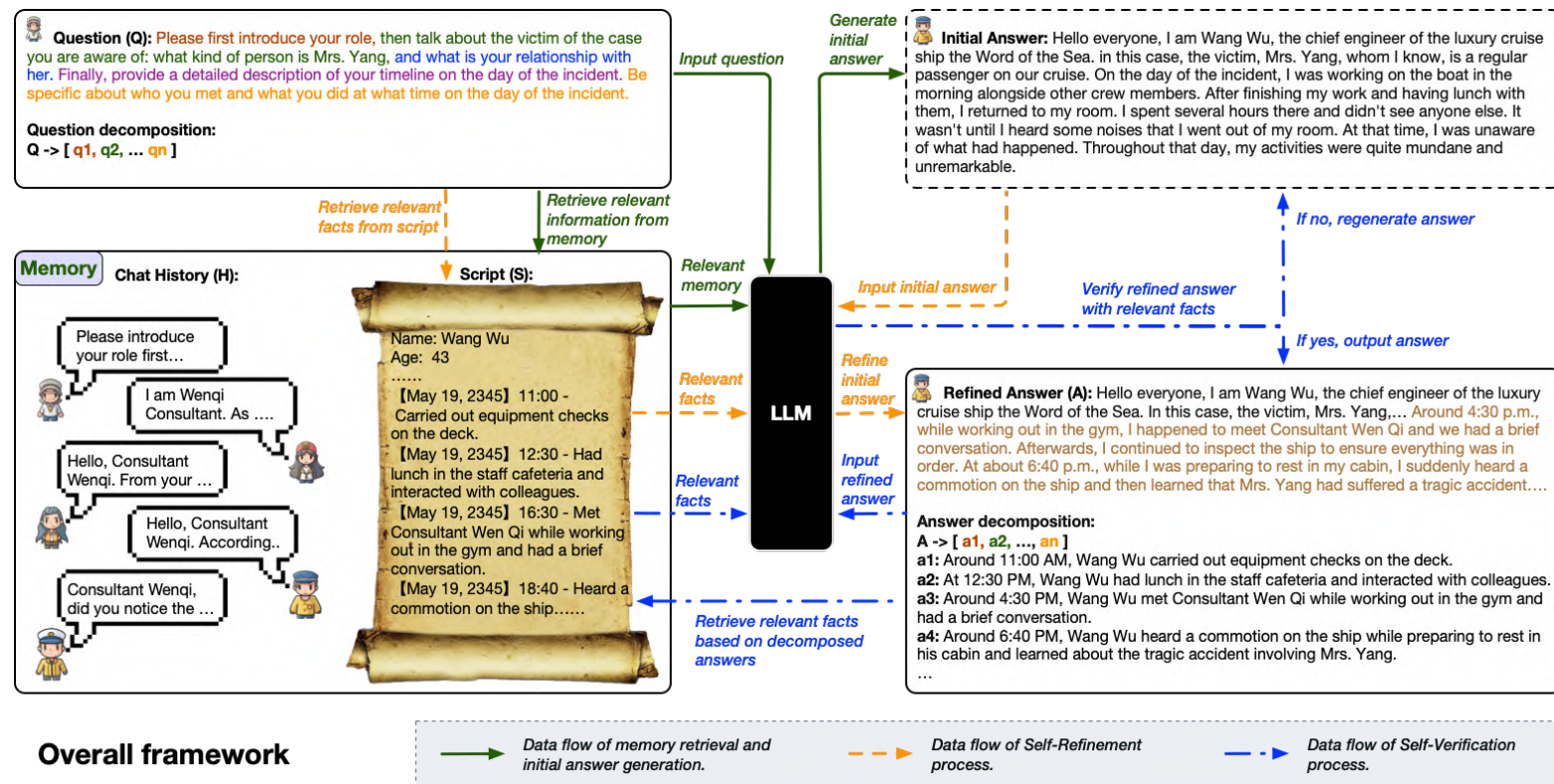
长程记忆和信息管理

协作与竞争的平衡

■ 目标：减少 Hallucination，增强 Reasoning



ThinkThrice: 三思而后行



数据集

- ❖ We collected 1,115 Jubensha game instances.
- ❖ Each game can have 1 to 20 players
- ❖ game token count can reach up to 518,000, enabling further research on socially intelligent AI.

数据申请与开源代码



项目网站: <https://jackwu502.github.io/ThinkThrice/>





Reset

Next

Stop Autoplay

Start Autoplay

实验设定

Example 1
Question: Who does Mate Zhang's brother work for? Answer: Boss Yang Source: He works for Boss Yang, owner of a famous bar in City A. (from Mate Zhang's script)
Example 2
Question: When did Dr. Ling meet Mate Zhang? Answer: 18:20 Source: 18:20, you went to meet Boss Yang, and met Mate Zhang just as you were about to arrive. (from Dr. Ling's script)

Question: Who is most likely to have made the thumping sound that Dr. Ling heard on her way to kill Boss Yang? A. Secretary Wen, B. Boss Yang, C. Boatman Wang, D. Mate Zhang, E. Others.
Answer: A. Secretary Wen
Rationale (GT): {Secretary Wen left Boss Yang's room through the ventilation duct at 23:00}PREMISE 1, and {Dr. Ling prepared to kill Boss Yang through the ventilation duct also at 23:00}PREMISE 2. Therefore, it can be inferred that {the 'thumping' sound Dr. Ling heard on her way to kill Boss Yang was most likely made by Secretary Wen}CONCLUSION.

Factual Questions

信息获取能力

Inferential Question

复杂推理能力

实验结果与观察

MR: Memory Retrieval
SR: Self-Refinement
SV: Self-Verification

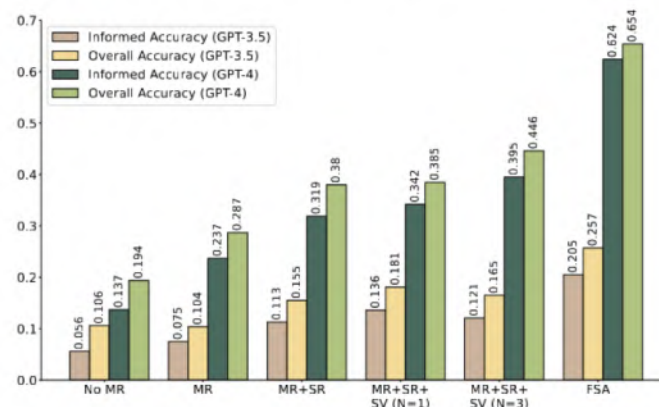
- ❖收集到的相关信息越多，代理通过推理解决问题的能力就越强。
- ❖在给定相同信息的情况下，LLM 的固有推理能力决定了代理解决问题的能力表现。
- ❖“三思”显著提升推理能力

Factual Question Answering

	Own Q		Other's Q	
	Avg	CQ	MQ	Avg
No MR	0.770	0.300	0.321	0.305
MR	0.759	0.409	0.380	0.402
MR+SR	0.757	0.467	0.487	0.471
MR+SR+SV(N=1)	0.768	0.485	0.518	0.492
MR+SR+SV(N=3)	0.772	0.495	0.514	0.498

	OpenAI	TF-IDF	Trigrams
No MR	0.062	0.000	0.000
MR	0.798	0.578	0.027
MR+SR	0.820	0.645	0.064
MR+SR+SV(N=1)	0.825	0.668	0.075
MR+SR+SV(N=3)	0.831	0.670	0.077

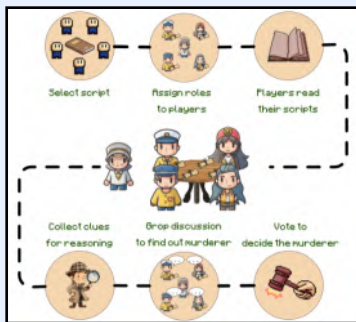
Inferential Question Answering



■ 小结：复杂推理Agent

Jubensha

剧本杀游戏的诸多特点使得其成为一个很好的评测任务，可以衡量Agent的多种能力



ThinkThrice

“三思”通过记忆检索，细节增强，反思修改来验证和加强推理结果，从而提高复杂推理的准确性，减少幻觉现象的影响。

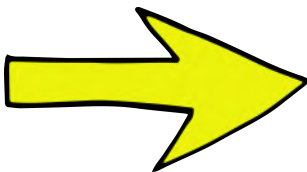


04 Scientific Agents

另一类复杂性：Knowledge

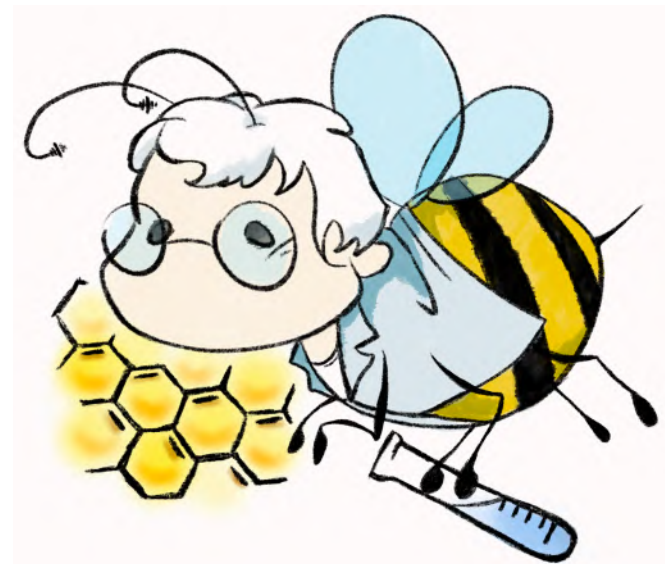
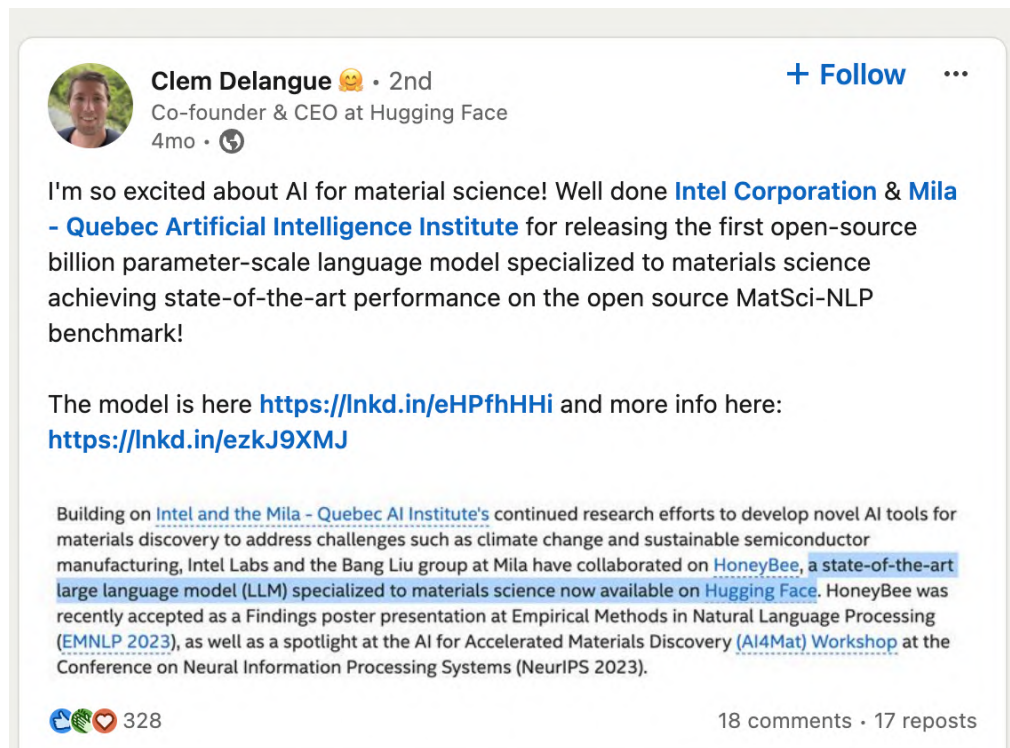


■ 如何让Agent获取专业知识，解决复杂任务？



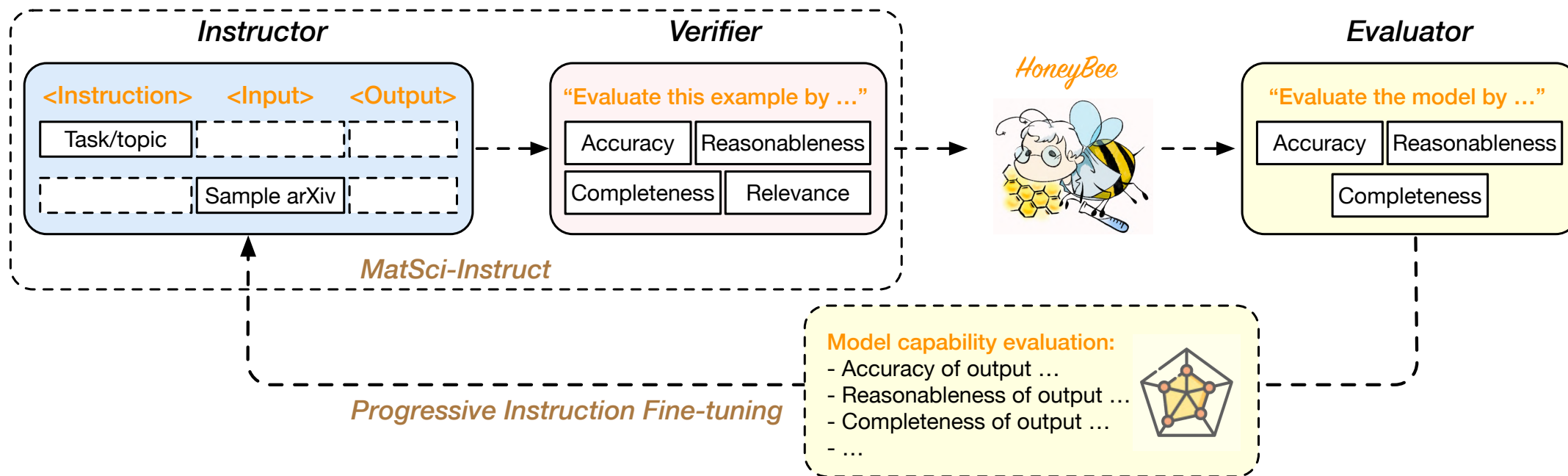
■ Key 1: 学习专业知识





HoneyBee is the first billion-parameter scale language model that is specialized in materials science

MatSci-Instruct and HoneyBee Training Workflow



Experimental Results

Model	Accuracy	Completeness	Reasonableness
Zero-Shot LLMs			
Chat-GPT	92.55	98.74	99.84
Llama-7b	78.81	90.36	97.64
Llama-13b	84.22	91.22	98.33
Alpaca-7b	81.35	92.01	98.49
Alpaca-13b	86.24	92.17	98.80
HoneyBee without Verification			
HoneyBee-7b	85.42	93.24	98.49
HoneyBee-13b	88.76	93.99	98.93
HoneyBee with MatSci-Instruct			
HB-7b-Stage1	88.81	93.42	99.07
HB-7b-Stage2	89.99	94.84	99.64
HB-7b-Stage3	91.95	95.78	99.90
HB-13b-Stage1	94.17	94.42	99.40
HB-13b-Stage2	96.42	95.42	99.78
HB-13b-Stage3	98.11	97.00	99.89

❖ Instruction data generated by our generator improves model performance

❖ The verifier part further improves data quality

❖ HoneyBee improves progressively

Experimental Results

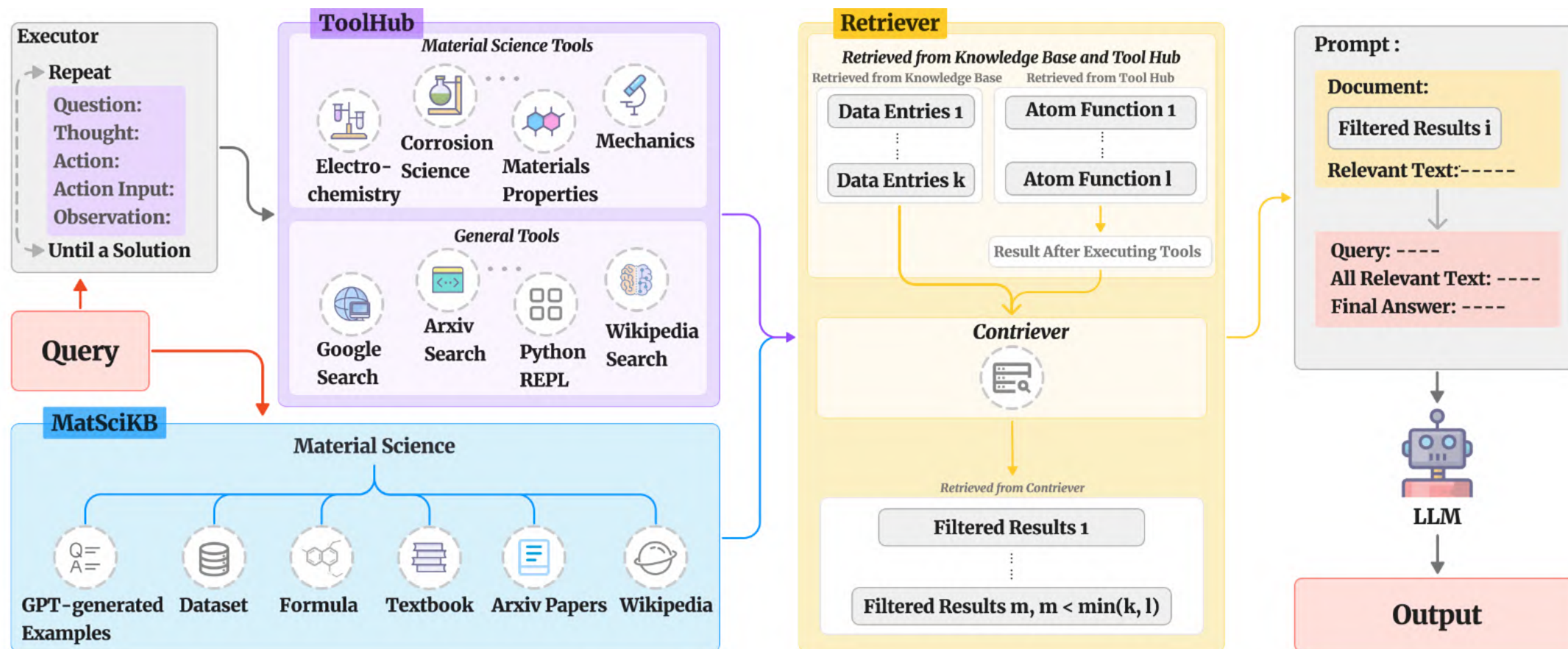
- ❖ We evaluate HoneyBee on MatSci-Instruct benchmark
- ❖ Low-resource setting: it out-performs BERT-family models
- ❖ Zero-Shot setting: it out-performs LLaMA-family models

Model	Named Entity Recognition	Relation Extraction	Event Argument Extraction	Paragraph Classification	Synthesis Action Retrieval	Sentence Classification	Slot Filling	Overall (All Tasks)
Low-Resource Finetuning on MatSci-NLP								
MatSciBERT (Gupta et al., 2022)	0.707	0.791	0.436	0.719	0.692	0.914	0.436	0.671
	0.470	0.507	0.251	0.623	0.484	0.660	0.194	0.456
MatBERT (Walker et al., 2021)	0.875	0.804	0.451	0.756	0.717	0.909	0.548	0.722
	0.630	0.513	0.288	0.691	0.594	0.614	0.273	0.517
HoneyBee-7b	0.787	0.852	0.551	0.741	0.792	0.991	0.529	0.749
	0.644	0.518	0.389	0.641	0.617	0.711	0.391	0.559
HoneyBee-13b	0.860	0.921	0.653	0.761	0.853	0.998	0.554	0.80
	0.748	0.578	0.486	0.658	0.662	0.743	0.401	0.611
Zero-Shot LLM Performance								
LLaMA-7b (Touvron et al., 2023)	0.042	0.094	0.160	0.279	0.052	0.096	0.142	0.208
	0.064	0.013	0.042	0.218	0.013	0.087	0.010	0.064
LLaMA-13b (Touvron et al., 2023)	0.057	0.109	0.042	0.233	0.039	0.079	0.138	0.1
	0.066	0.016	0.054	0.189	0.009	0.074	0.008	0.059
Alpaca-7b (Taori et al., 2023)	0.031	0.053	0.029	0.375	0.179	0.180	0.139	0.141
	0.018	0.037	0.009	0.294	0.129	0.180	0.039	0.101
Alpaca-13b (Taori et al., 2023)	0.053	0.016	0.111	0.310	0.442	0.375	0.110	0.202
	0.046	0.035	0.072	0.237	0.278	0.334	0.015	0.145
Chat-GPT (OpenAI, 2022)	0.063	0.232	0.204	0.433	0.300	0.320	0.368	0.274
	0.052	0.145	0.203	0.450	0.183	0.318	0.280	0.233
Claude (Bai et al., 2022)	0.063	0.232	0.195	0.442	0.280	0.329	0.393	0.276
	0.048	0.143	0.169	0.467	0.177	0.326	0.305	0.234
GPT-4 (OpenAI, 2023)	0.189	0.445	0.453	0.679	0.743	0.788	0.502	0.543
	0.121	0.432	0.353	0.522	0.677	0.689	0.483	0.468
Zero-Shot HoneyBee with MatSci-Instruct								
HoneyBee-7b-Stage1	0.173	0.138	0.196	0.380	0.592	0.416	0.292	0.301
	0.148	0.120	0.096	0.207	0.208	0.334	0.105	0.174
HoneyBee-7b-Stage2	0.243	0.199	0.237	0.440	0.612	0.467	0.344	0.363
	0.166	0.145	0.123	0.301	0.289	0.345	0.176	0.221
HoneyBee-7b-Stage3	0.267	0.245	0.290	0.490	0.688	0.490	0.393	0.409
	0.190	0.178	0.189	0.343	0.342	0.365	0.289	0.271
HoneyBee-13b-Stage1	0.369	0.301	0.389	0.500	0.701	0.512	0.467	0.463
	0.256	0.224	0.265	0.379	0.378	0.402	0.334	0.320
HoneyBee-13b-Stage2	0.391	0.367	0.437	0.576	0.765	0.557	0.508	0.514
	0.299	0.290	0.303	0.411	0.401	0.461	0.379	0.363
HoneyBee-13b-Stage3	0.429	0.412	0.481	0.611	0.801	0.589	0.578	0.557
	0.372	0.346	0.378	0.467	0.429	0.503	0.423	0.417

■ Key 2: 善用外部工具



HoneyComb



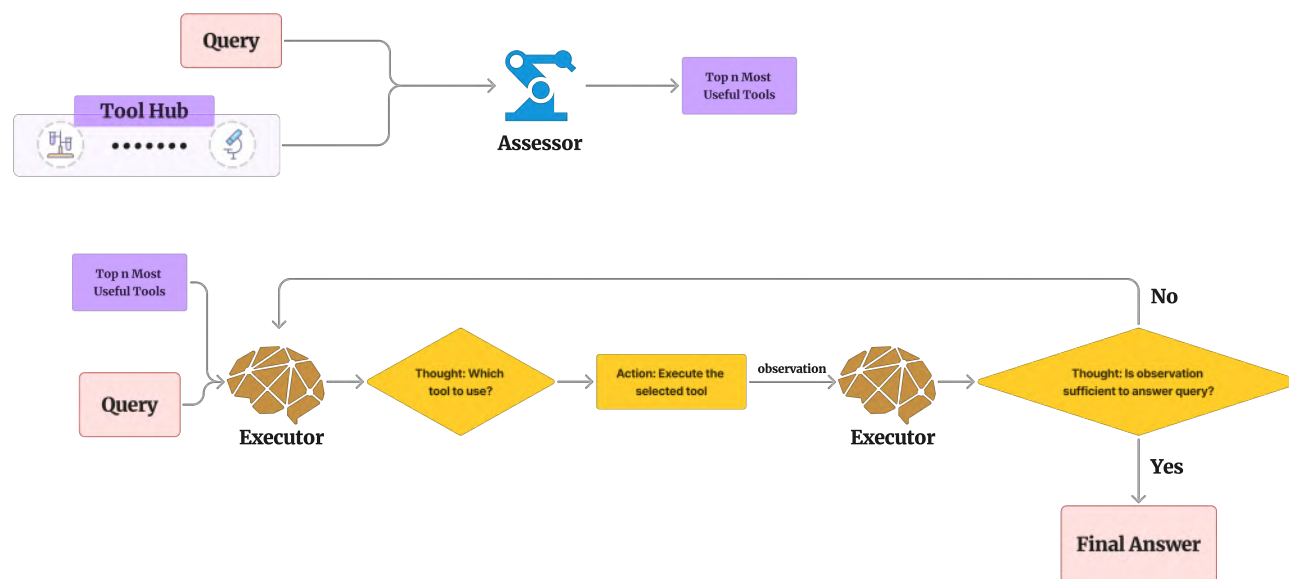
工具构造和使用

Algorithm 1 Inductive Tool Construction

Require: Train Set D_{train} , LLM M

Ensure: Set of atom tools A

- 1: $A \leftarrow \emptyset$ {Initialize the set of atom functions}
- 2: **for** each question q_i in D_{train} **do**
- 3: $f_i \leftarrow M(q_i)$ {Generate specific function for q_i }
- 4: Human verifies f_i
- 5: Decompose f_i into atom functions a_i
- 6: $A \leftarrow A \cup a_i$ {Add atom functions to the set}
- 7: **end for**
- 8: **return** A



实验效果

Dataset	HoneyBee	HoneyBee 🐝	GPT-3.5	GPT-3.5 + 🐝	GPT-4	GPT-4 + 🐝	Llama2	Llama2 + 🐝	Llama3	Llama3 + 🐝
MaScQA	16.62	33.38	33.54	38.46	58.46	79.07	22.15	36.31	24.62	47.23
SciQA	33.96	79.69	90.69	90.83	90.84	96.54	75.79	78.66	93.00	93.32

Table 3: HoneyComb evaluation with diverse LLMs including open-source LLMs (HoneyBee (Song et al., 2023b), LLaMa2 (Touvron et al., 2023b), LLaMa3 (AI@Meta, 2024)) and commercial LLMs (GPT3.5, GPT4 (OpenAI, 2024)). The results show that HoneyComb consistently improves the performance of all LLMs for SciQA and MaScQA.

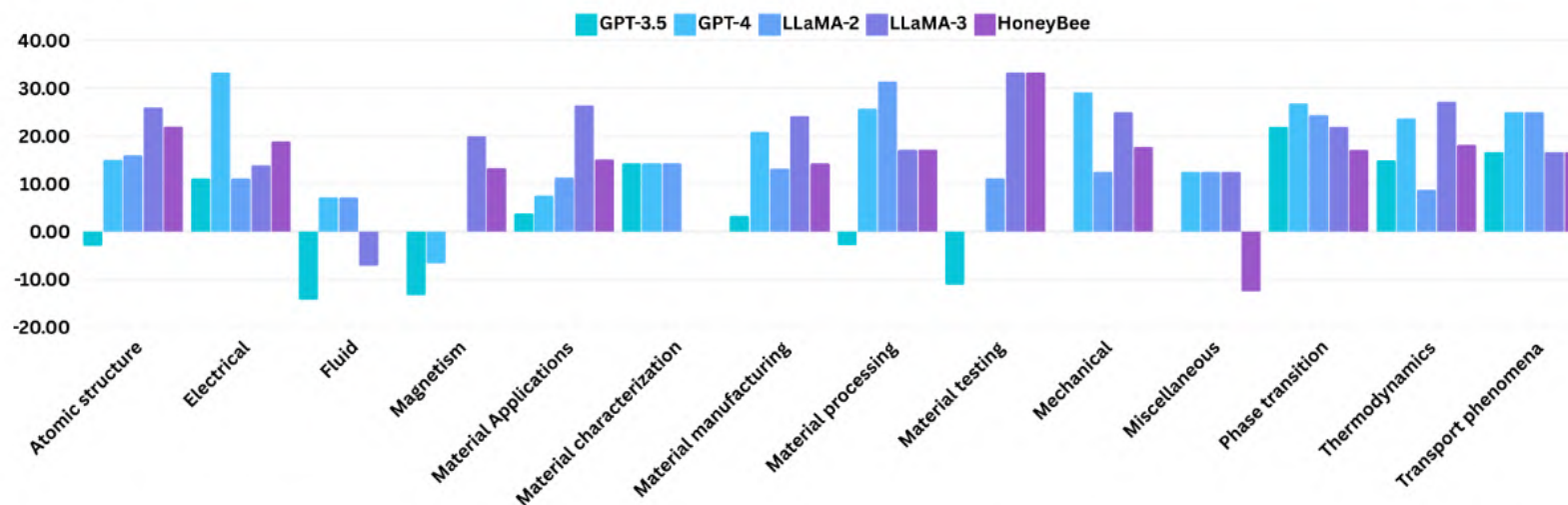


Figure 3: Improvements of various LLMs integrated with HoneyComb compared to relevant baseline LLMs for different materials science tasks. With few exceptions, HoneyComb improves the performance of all LLMs across all tasks showing the utility of tool augmentation.

❖所有基于 HoneyComb 的模型在 MaScQA 和 SciQA 上的准确率均有显著提升。

❖总体趋势表明，HoneyComb 显著提高了模型性能。LLaMA-3 和 HoneyBee 有明显提升。

■ Key 3: 化繁为简, 统筹思考



■ Data Interpreter: 智能体也能玩转数据

DATA INTERPRETER: AN LLM AGENT FOR DATA SCIENCE

**Sirui Hong^{1*}, Yizhang Lin^{1*}, Bang Liu^{2†}, Bangbang Liu^{1†}, Binhao Wu^{1†},
Danyang Li^{1†}, Jiaqi Chen^{3†}, Jiayi Zhang^{4†}, Jinlin Wang^{1†}, Li Zhang^{3†}, Lingyao Zhang[†],
Min Yang^{5†}, Mingchen Zhuge^{6†}, Taicheng Guo^{7†}, Tuo Zhou^{8†}, Wei Tao^{3†},
Wenyi Wang^{6†}, Xiangru Tang^{9†}, Xiangtao Lu^{1†}, Xiawu Zheng^{10†}, Xinbing Liang^{1,11†},
Yaying Fei^{12†}, Yuheng Cheng^{13†}, Zongze Xu^{1,14†}, Chenglin Wu^{1†}**

¹DeepWisdom, ²Université de Montréal & Mila, ³Fudan University ⁴Renmin University of China

⁵Shenzhen Institute of Advanced Technology, Chinese Academy of Sciences

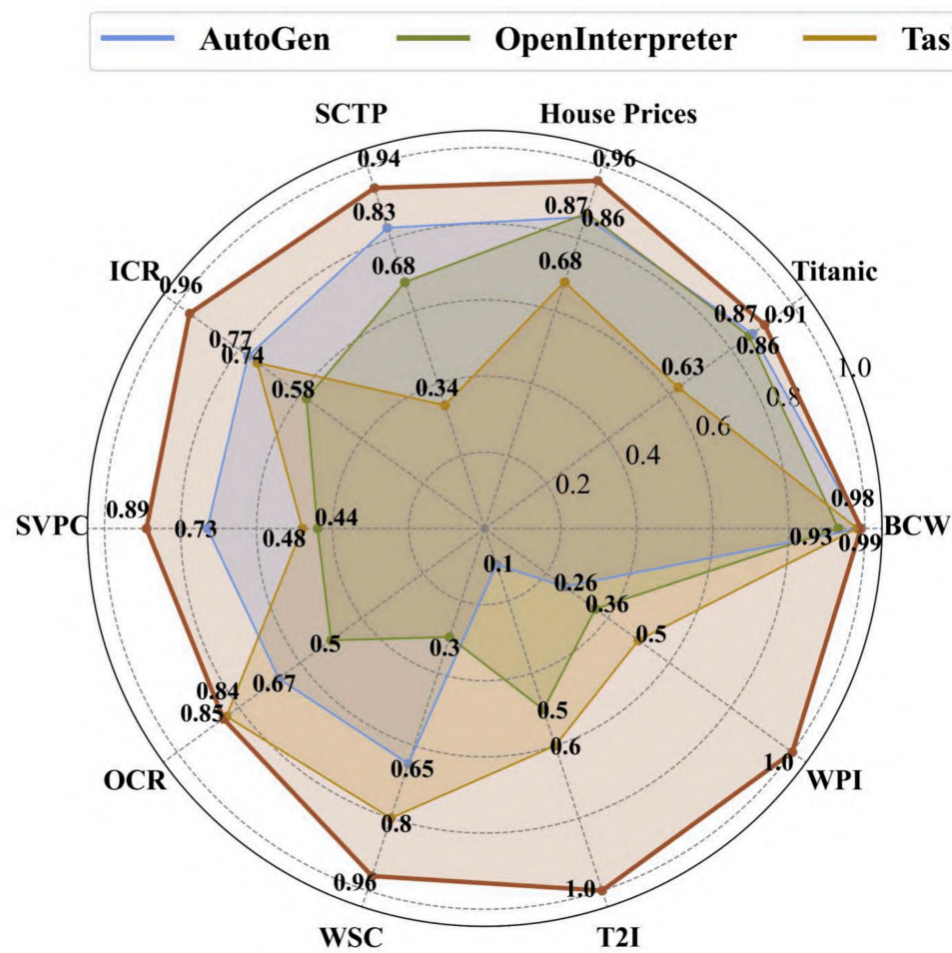
⁶AI Initiative, King Abdullah University of Science and Technology

⁷University of Notre Dame ⁸The University of Hong Kong ⁹Yale University

¹⁰Xiamen University ¹¹East China Normal University ¹²Beijing University of Technology

¹³The Chinese University of Hong Kong, Shenzhen ¹⁴Hohai University

Data Interpreter: 智能体也能玩转数据



Machine learning tasks

- SVPC: Santander Value Prediction Challenge
- ICR: Identifying Age-Related Conditions
- Sctp: Santander Customer Transaction Prediction
- BCW: Breast Cancer Wisconsin
- House Prices Prediction
- Titanic Survival Prediction

Real-world open-ended tasks

- OCR: Optical Character Recognition
- WSC: Web Search and Crawling
- T2I: Text-to-Image
- WPI: Web Page Imitation

数据分析可视化

```
File Edit Selection View ... MetaGPT
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS python + v [] [] ... v x

    "task_type": "model train"
  },
  {
    "task_id": "5",
    "dependent_task_ids": ["4"],
    "instruction": "Evaluate the trained models using appropriate metrics to select the best performing model.",
    "task_type": "model evaluate"
  },
  {
    "task_id": "6",
    "dependent_task_ids": ["5"],
    "instruction": "Apply the best performing model to the test.csv data to predict production and update the test.csv with the predictions.",
    "task_type": "other"
  }
]
...

2024-04-07 15:20:16.743 | INFO      | metagpt.utils.cost_manager:update_cost:57 - Total running cost: $0.016 | Max budget: $10.000
| Current cost: $0.016, prompt_tokens: 608, completion_tokens: 339
2024-04-07 15:20:16.744 | INFO      | metagpt.roles.role:plan_and_act:486 - ready to take on task task_id='1' dependent_task_ids=
[] instruction='Perform exploratory data analysis on the provided train.csv to understand the data distribution, missing values,
and potential correlations.' task_type='eda' code='' result='' is_success=False is_finished=False
2024-04-07 15:20:16.746 | INFO      | metagpt.tools.tool_recommend:recall_tools:195 - Recalled tools:
['FillMissingValue', 'GPTvGenerator', 'SDEngine', 'StandardScale', 'OneHotEncode', 'RobustScale', 'VarianceBasedSelection', 'TargetMeanEncoder', 'KFoldTargetMeanEncoder', 'MinMaxScale', 'email_login_imap', 'scrape_web_playwright', 'LabelEncode', 'CatCross',
'SplitBins', 'PolynomialExpansion', 'MaxAbsScale', 'OrdinalEncode', 'GeneralSelection', 'CatCount']; Scores: [29.677522492793837,
26.012622975391647, 25.291876124154793, 25.282965930471164, 25.154574432521414, 25.12759216705134, 24.919798631899567, 24.907845
875436156, 24.419822564719382, 24.407909060645736, 23.93769136967075, 23.91939432961399, 23.86256684139465, 23.69006174582066, 23
.6859173879769, 23.66551020109199, 23.403191954448154, 23.354521008586477, 23.10902635512132, 22.80301033004188]
```


异常数据处理

```
File Edit Selection View ... MetaGPT
```

	PROBLEMS	OUTPUT	DEBUG CONSOLE	TERMINAL	PORTS
25%	-1.027461	-2.213298	-1.628854	-4.252406	0.055267
50%	0.838357	0.270344	1.527967	0.687264	0.204210
75%	2.550258	2.029854	4.617410	3.658531	0.500000
max	11.117029	9.508798	16.988418	14.314416	0.999530
	u10	v10	u100	v100	production
u10	1.000000	-0.051057	0.979519	-0.016381	0.318136
v10	-0.051057	1.000000	-0.091770	0.979383	-0.128665
u100	0.979519	-0.091770	1.000000	-0.058405	0.307901
v100	-0.016381	0.979383	-0.058405	1.000000	-0.143718
production	0.318136	-0.128665	0.307901	-0.143718	1.000000

```
*****  
2024-04-07 15:21:07.478 | INFO      | metagpt.roles.role:_plan_and_act:486 - ready to take on task task_id='2' dependent_task_ids=  
['1'] instruction='Preprocess the dataset by handling missing values, scaling features, and encoding categorical variables if any  
' task_type='data preprocessing' code='' result='' is_success=False is_finished=False  
2024-04-07 15:21:07.480 | INFO      | metagpt.tools.tool_recommend:recall_tools:195 - Recalled tools:  
['FillMissingValue', 'StandardScale', 'MinMaxScale', 'KFoldTargetMeanEncoder', 'OneHotEncode', 'TargetMeanEncoder', 'LabelEncode'  
, 'OrdinalEncode', 'GroupStat', 'GPtvGenerator', 'CatCross', 'VarianceBasedSelection', 'RobustScale', 'MaxAbsScale', 'CatCount',  
'PolynomialExpansion', 'scrape_web_playwright', 'SplitBins', 'GeneralSelection', 'SDEngine']; Scores: [24.823028844453535, 23.101  
08131089997, 22.586821914971672, 22.30150829131259, 22.010333589978888, 21.62872720696448, 21.346009048923037, 20.11001327345078,  
19.893290184569462, 19.67712816930853, 19.623384765206488, 19.616970058644768, 19.59278538866703, 19.577200988642733, 19.5721329  
47552262, 19.40874077006594, 19.335950251392944, 18.957584180164012, 18.915760497652183, 18.77332318310233]  
``json  
[\"FillMissingValue\", \"StandardScale\", \"MinMaxScale\", \"OneHotEncode\", \"LabelEncode\"]  
...  
2024-04-07 15:21:09.989 | INFO      | metagpt.utils.cost_manager:update_cost:57 - Total running cost: $0.007 | Max budget: $10.000  
| Current cost: $0.007, prompt_tokens: 621, completion_tokens: 26  
2024-04-07 15:21:09.990 | INFO      | metagpt.tools.tool_recommend:recommend_tools:101 - Recommended tools:  
['FillMissingValue', 'StandardScale', 'MinMaxScale', 'OneHotEncode', 'LabelEncode']  
2024-04-07 15:21:09.991 | INFO      | metagpt.roles.di.data_interpreter:_check_data:180 - Check updated data
```


自动化特征工程

```
File Edit Selection View ... MetaGPT
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS python + - x

    u10      v10      u100      v100 year month day hour
0  0.569407  0.578869  0.539713  0.573027 2021    1    1    0
1  0.588368  0.604705  0.556854  0.595244 2021    1    1    1
2  0.602349  0.609532  0.569849  0.598813 2021    1    1    2
3  0.619189  0.625000  0.586157  0.613004 2021    1    1    3
4  0.628392  0.635599  0.595503  0.622844 2021    1    1    4

2024-04-07 15:22:05.996 | INFO      | metagpt.roles.role:plan_and_act:486 - ready to take on task task_id='3' dependent_task_ids=
['2'] instruction='Create new features that could improve model performance, such as wind speed and direction from u10, v10, u100
, and v100.' task_type='feature engineering' code='' result='' is_success=False is_finished=False
2024-04-07 15:22:05.997 | INFO      | metagpt.tools.tool_recommend:recall_tools:195 - Recalled tools:
['GPTvGenerator', 'TargetMeanEncoder', 'CatCount', 'PolynomialExpansion', 'RobustScale', 'OrdinalEncode', 'SEngine', 'MinMaxScale',
'GroupStat', 'CatCross', 'VarianceBasedSelection', 'StandardScale', 'KFoldTargetMeanEncoder', 'OneHotEncode', 'SplitBins', 'GeneralSelection',
'scrape_web_playwright', 'email_login_imap', 'MaxAbsScale', 'FillMissingValue']; Scores: [30.427133913989564, 2
9.87038301676159, 29.149291656609687, 29.061885220858986, 28.418202517964776, 28.268685886449582, 27.644138817364425, 27.43379707
2806218, 27.209011501270506, 27.114415287898716, 27.094464370358875, 26.89637909890323, 26.867246664923393, 26.523351555680637, 2
6.52222750559041, 26.39325480936629, 26.350841514275093, 25.949518542061284, 25.76811556674519, 25.696367969881155]
```json
["PolynomialExpansion"]
```

2024-04-07 15:22:07.584 | INFO      | metagpt.utils.cost_manager:update_cost:57 - Total running cost: $0.007 | Max budget: $10.000
| Current cost: $0.007, prompt_tokens: 651, completion_tokens: 9
2024-04-07 15:22:07.585 | INFO      | metagpt.tools.tool_recommend:recommend_tools:101 - Recommended tools:
['PolynomialExpansion']
2024-04-07 15:22:07.587 | INFO      | metagpt.roles.di.data_interpreter:_check_data:180 - Check updated data
```python
from metagpt.tools.libs.data_preprocess import get_column_info

Since the last operation in the finished tasks was on train_df_processed and test_df_processed,

```

# 自动化特征工程

```
File Edit Selection View ... MetaGPT
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS python + - x

wind_speed_10 wind_speed_100 wind_speed_10 wind_dir_100 wind_dir_10^2 \
0 0.402824 0.429037 0.393860
1 0.457478 0.480272 0.439616
2 0.506232 0.536141 0.505208
3 0.525760 0.576399 0.572889
4 0.541278 0.608704 0.630411

wind_dir_10 wind_speed_100 wind_dir_10 wind_dir_100 wind_speed_100^2 \
0 0.395939 0.421705 0.398029
1 0.444210 0.466342 0.448851
2 0.499202 0.528696 0.493267
3 0.541025 0.593134 0.510933
4 0.575661 0.647370 0.525666

wind_speed_100 wind_dir_100 wind_dir_100^2
0 0.423931 0.451518
1 0.471215 0.494693
2

2024-04-07 15:22:25.836 | INFO | metagpt.roles.role:plan_and_act:486 - ready to take on task task_id='4' dependent_task_ids=
['3'] instruction='Train different machine learning models using the preprocessed and feature-engineered data to predict wind far
m production.' task_type='model train' code='' result='' is_success=False is_finished=False
2024-04-07 15:22:25.838 | INFO | metagpt.tools.tool_recommend:recall_tools:195 - Recalled tools:
['KFoldTargetMeanEncoder', 'RobustScale', 'OneHotEncode', 'LabelEncode', 'SplitBins', 'TargetMeanEncoder', 'StandardScale', 'CatC
ross', 'PolynomialExpansion', 'MinMaxScale', 'SDEngine', 'scrape_web_playwright', 'VarianceBasedSelection', 'GeneralSelection', '
MaxAbsScale', 'CatCount', 'email_login_imap', 'GPTvGenerator', 'OrdinalEncode', 'FillMissingValue']; Scores: [24.079782287730822,
22.802608313913375, 22.385748944870794, 22.1114139699989, 21.946888770688528, 21.500534156731394, 21.495613186689262, 21.2814576
80746062, 21.244885622340433, 21.01893832621634, 20.956780924706585, 20.844234845748495, 20.694026245631672, 20.694026245631672,
20.692823543104215, 20.618715560856323, 20.43444493282982, 20.367195985387983, 20.017287083445318, 20.017287083445318]
```



# 模型训练及验证

```
File Edit Selection View ... MetaGPT
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS python + - ... - x

4 # we will print the column information for this DataFrame.
5 column_info_test_df = get_column_info(test_df_expanded)
6 print("column_info_test_df")
7 print(column_info_test_df)
8
9 # Additionally, we will print the column information for train_df_expanded as well,
10 # as it is the corresponding training DataFrame.
11 column_info_train_df = get_column_info(train_df_expanded)
12 print("column_info_train_df")
13 print(column_info_train_df)
14
column_info_test_df
{'Category': [], 'Numeric': ['year', 'month', 'day', 'hour', 'u10', 'v10', 'u100', 'v100', 'wind_speed_10', 'wind_dir_10', 'wind_speed_100', 'wind_dir_100', 'u10^2', 'u10 v10', 'u10 u100', 'u10 v100', 'u10 wind_speed_10', 'u10 wind_dir_10', 'u10 wind_speed_100', 'u10 wind_dir_100', 'v10^2', 'v10 u100', 'v10 v100', 'v10 wind_speed_10', 'v10 wind_dir_10', 'v10 wind_speed_100', 'v10 wind_dir_100', 'u100^2', 'u100 v100', 'u100 wind_speed_10', 'u100 wind_dir_10', 'u100 wind_speed_100', 'u100 wind_dir_100', 'v100^2', 'v100 wind_speed_10', 'v100 wind_dir_10', 'v100 wind_speed_100', 'v100 wind_dir_100', 'wind_speed_10^2', 'wind_speed_10 wind_dir_10', 'wind_speed_10 wind_speed_100', 'wind_speed_10 wind_dir_100', 'wind_dir_10^2', 'wind_dir_10 wind_speed_100', 'wind_dir_10 wind_dir_100', 'wind_speed_100^2', 'wind_speed_100 wind_dir_100', 'wind_dir_100^2'], 'Datetime': [], 'Others': []}
column_info_train_df
{'Category': [], 'Numeric': ['production', 'year', 'month', 'day', 'hour', 'u10', 'v10', 'u100', 'v100', 'wind_speed_10', 'wind_dir_10', 'wind_speed_100', 'wind_dir_100', 'u10^2', 'u10 v10', 'u10 u100', 'u10 v100', 'u10 wind_speed_10', 'u10 wind_dir_10', 'u10 wind_speed_100', 'u10 wind_dir_100', 'v10^2', 'v10 u100', 'v10 v100', 'v10 wind_speed_10', 'v10 wind_dir_10', 'v10 wind_speed_100', 'v10 wind_dir_100', 'u100^2', 'u100 v100', 'u100 wind_speed_10', 'u100 wind_dir_10', 'u100 wind_speed_100', 'u100 wind_dir_100', 'v100^2', 'v100 wind_speed_10', 'v100 wind_dir_10', 'v100 wind_speed_100', 'v100 wind_dir_100', 'wind_speed_10^2', 'wind_speed_10 wind_dir_10', 'wind_speed_10 wind_speed_100', 'wind_speed_10 wind_dir_100', 'wind_dir_10^2', 'wind_dir_10 wind_speed_100', 'wind_dir_10 wind_dir_100', 'wind_speed_100^2', 'wind_speed_100 wind_dir_100', 'wind_dir_100^2'], 'Datetime': [], 'Others': []}

2024-04-07 15:22:35.344 | INFO | metagpt.roles.di.data_interpreter:write_code:153 - ready to WriteAnalysisCode
```

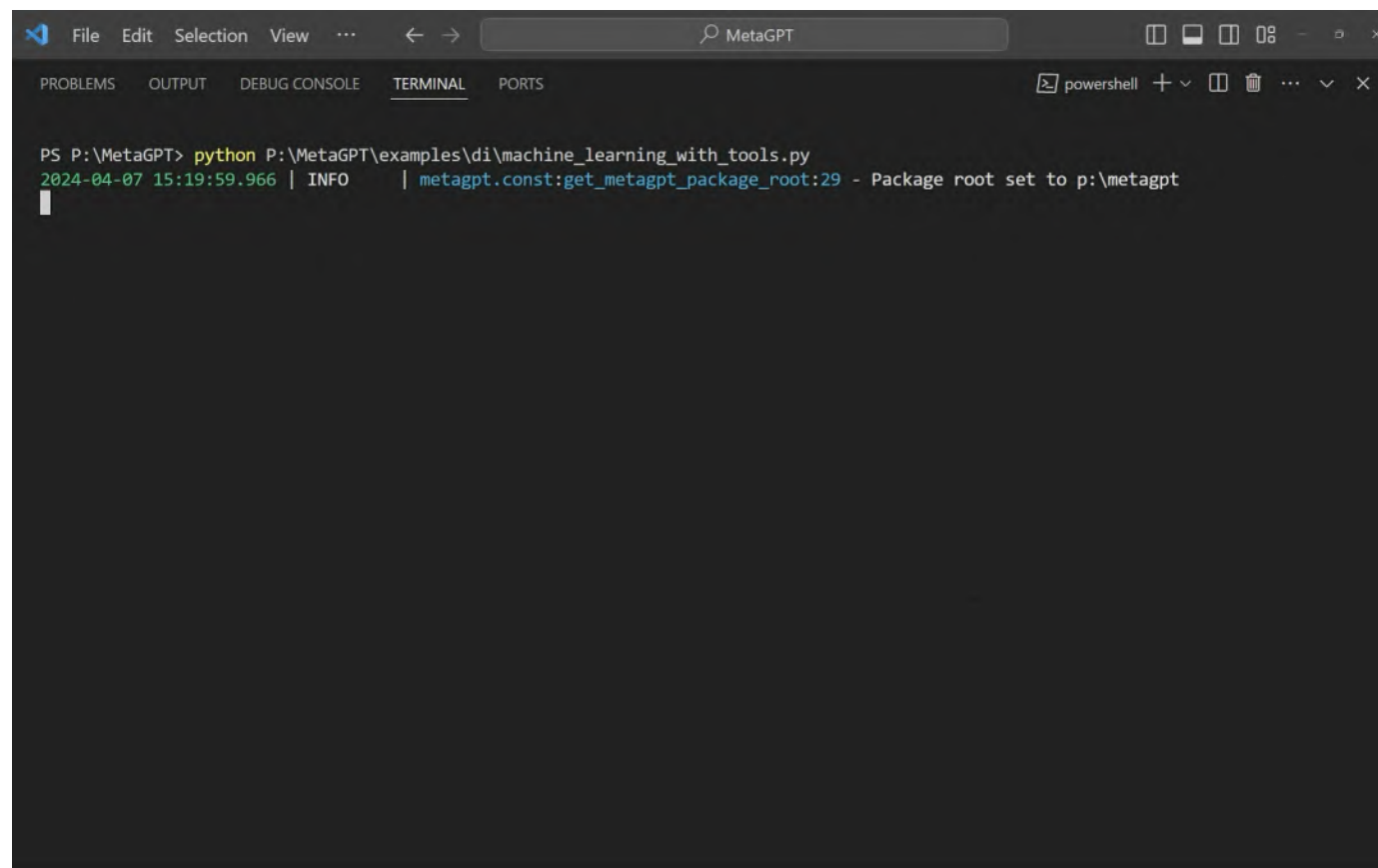


# 任务规划不可控 – 任务规划 (DAG)

❖ **问题：** LLM在数据科学任务中，面对实时数据变化、复杂依赖关系和逻辑错误检测的需求，现有计划生成方法难以满足

❖ **分析：** 静态规划不适合动态数据依赖场景，且逻辑严谨性难以保障，需灵活适应任务变化及高效管理依赖关系

❖ **解法：** 采用动态层次规划，构建任务行动图以适应数据变化，动态更新任务状态和代码，结合自我调试和人工编辑策略纠正逻辑错误，通过信心评分和验证提升推理准确性

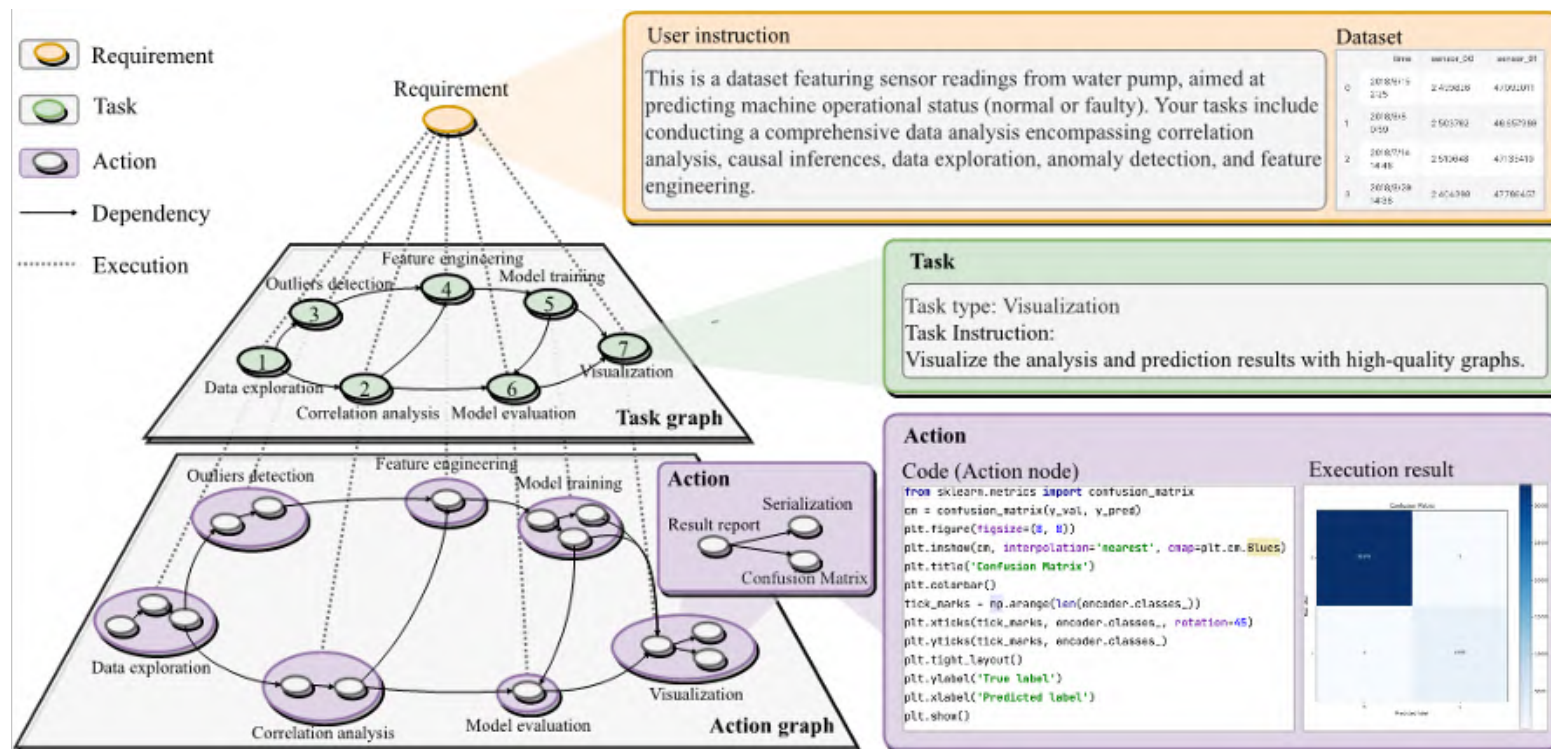


```
File Edit Selection View ... MetaGPT
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS P:\MetaGPT> python P:\MetaGPT\examples\di\machine_learning_with_tools.py
2024-04-07 15:19:59.966 | INFO | metagpt.const:get_metagpt_package_root:29 - Package root set to p:\metagpt
```

# Hierarchical Graph: 化繁为简，层次管理

分层结构：(a) 一个有组织的**任务和动作图**，展示了高层级机器学习项目的工作流程，包括实现项目目标所需的任务依赖和动作序列。

(b) **任务的有向无环图 (DAG)**，以机器操作状态预测问题为例。任务图展示了拆解的计划任务，而动作图（也称为执行图）则根据计划的执行代码由LLM转换

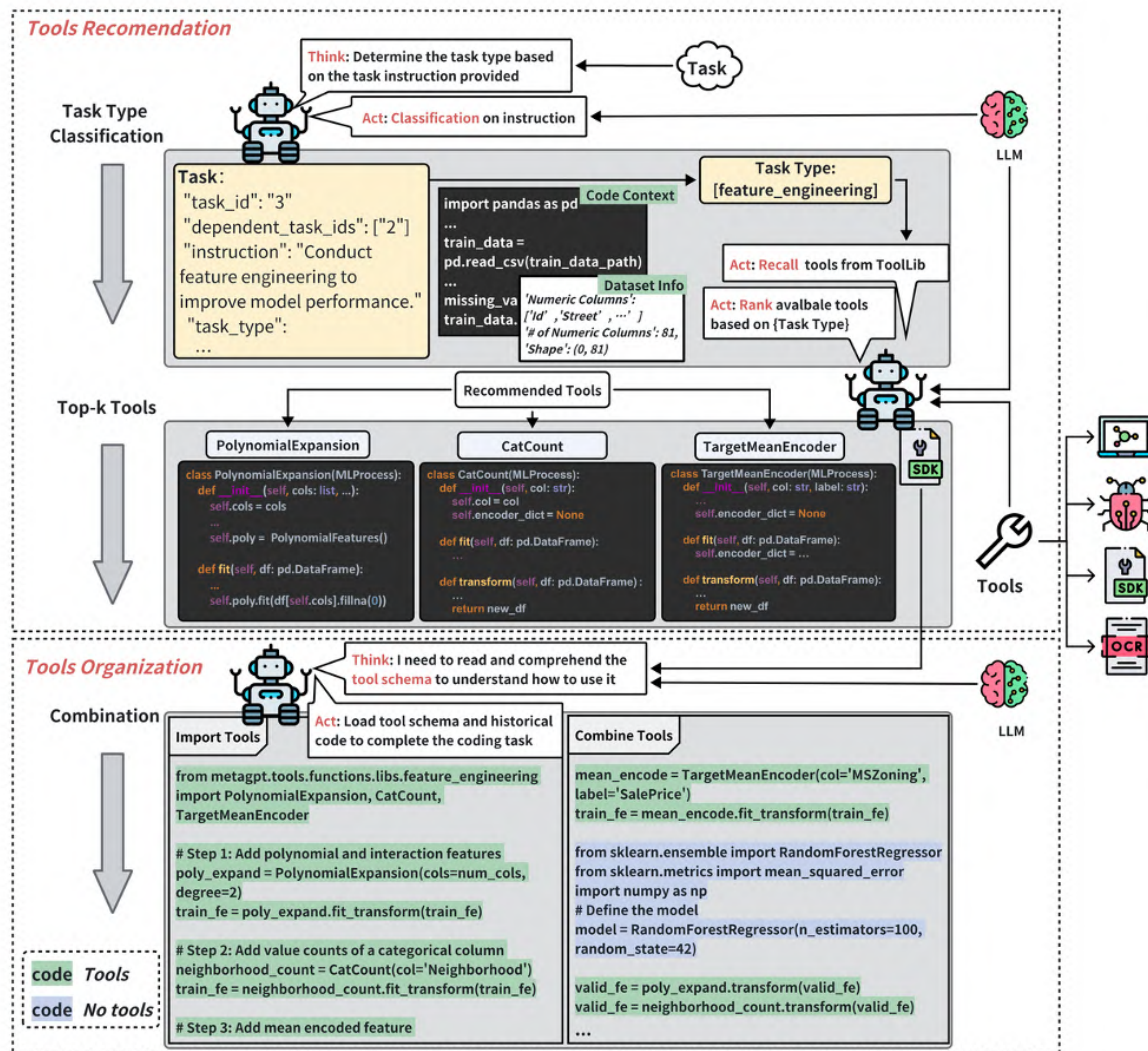


# 代码生成不可控 – 工具集成与生成

❖ **问题：** 复杂数据科学任务中，手工编写工具不易，需有效整合与自动生成工具

❖ **分析：** 直接通过LLM生成复杂工具代码有挑战，且现有工具推荐和组织方式效率低

❖ **解法：** Data Interpreter动态推荐和组织工具，利用LLM理解工具功能并嵌入上下文调整参数，整合多种工具并自动化生成代码片段，通过执行反馈迭代优化工具库，强化逻辑一致性和代码效率





# 代码生成不可控 – 工具集成与生成

## Tools of Data Interpreter

❖ **问题：**复杂数据科学任务中，手工编写工具不易，需有效整合与自动生成工具

❖ **分析：**直接通过LLM生成复杂工具代码有挑战，且现有工具推荐和组织方式效率低

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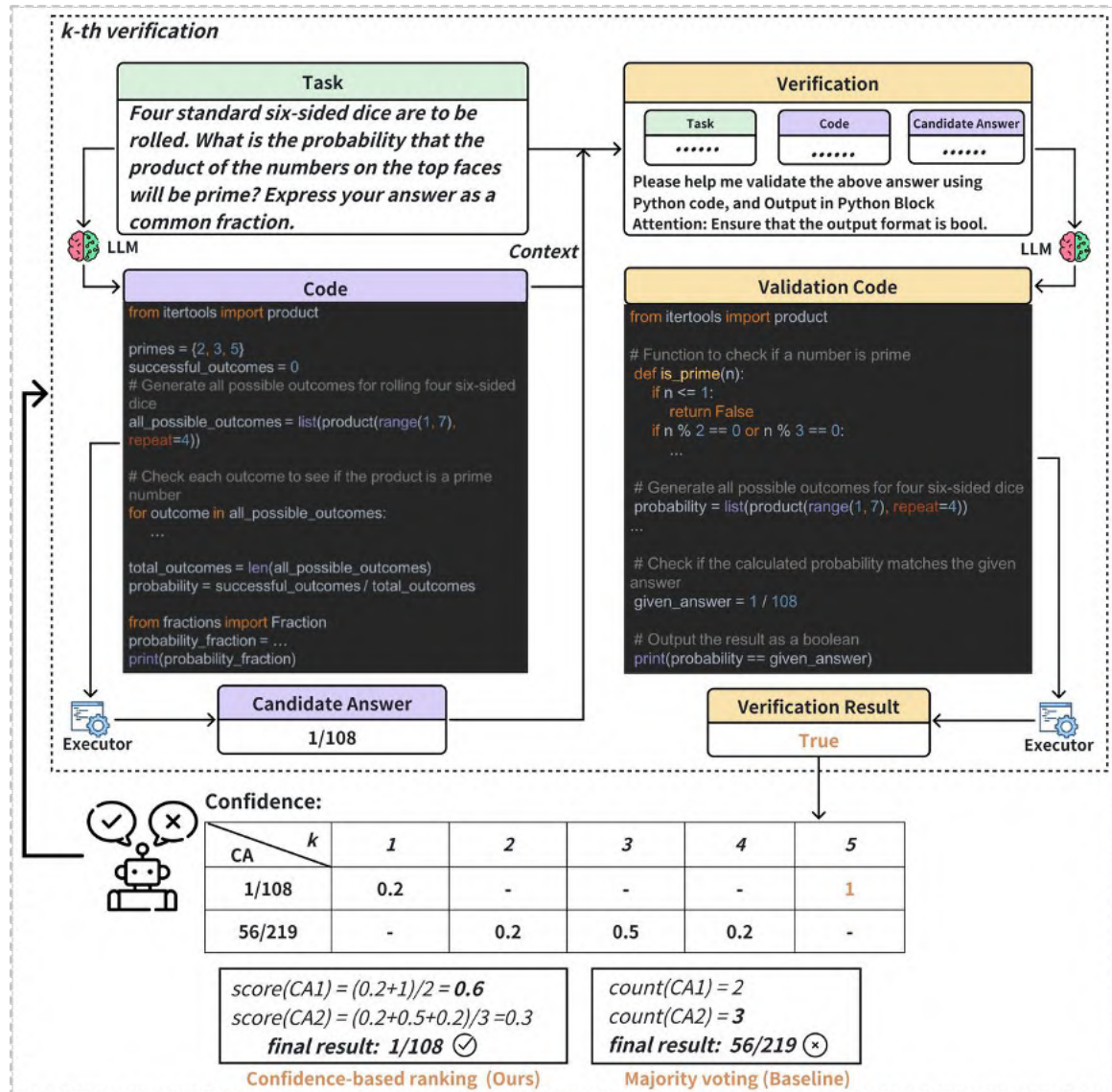
Tool name	Tool type	Functions	Domain
FillMissingValue	Class	4	Machine learning
MinMaxScale	Class	4	Machine learning
StandardScale	Class	4	Machine learning
MaxAbsScale	Class	4	Machine learning
LabelEncode	Class	4	Machine learning
OneHotEncode	Class	4	Machine learning
OrdinalEncode	Class	4	Machine learning
RobustScale	Class	4	Machine learning
CatCount	Class	4	Machine learning
TargetMeanEncoder	Class	4	Machine learning
KFoldTargetMeanEncoder	Class	4	Machine learning
CatCross	Class	5	Machine learning
SplitBins	Class	4	Machine learning
GeneralSelection	Class	4	Machine learning
TreeBasedSelection	Class	4	Machine learning
VarianceBasedSelection	Class	4	Machine learning
PolynomialExpansion	Class	4	Machine learning
GPTvGenerator	Class	3	Multimodal
SDEngine	Class	5	Multimodal
scrape_web_playwright	Function	1	Common

# 代码生成不可控 – 自动验证

❖ **问题：**即便代码无误，也可能存在逻辑缺陷，需有效验证代码输出的正确性

❖ **分析：**单纯依赖异常检测不能确保任务完成，需对逻辑严密性进行深入验证

❖ **解法：**Data Interpreter采用ACV (Automated Confidence-based Verification) 技术，生成验证代码模拟任务逻辑过程，依据执行结果计算置信分数，据此选择最可靠答案，有效避免逻辑错误，提高推理准确性



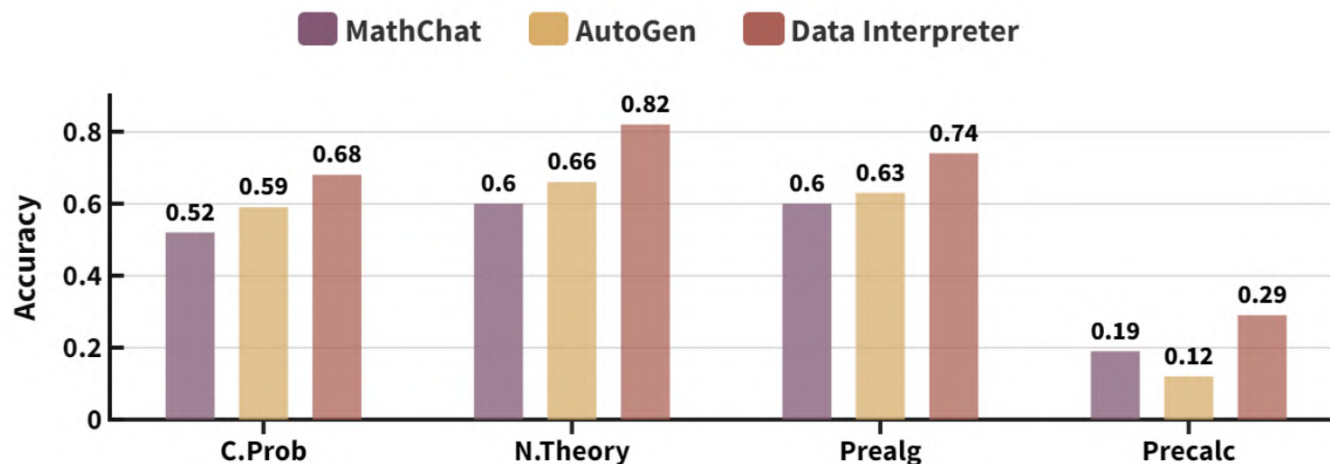
以MATH内的一个任务说明基于置信度自动验证流程：虚线框内是自动验证的过程，虚线框下方根据验证对多个候选答案进行排序

# 代码生成不可控 – 自动验证

❖ **问题：**即便代码无误，也可能存在逻辑缺陷，需有效验证代码输出的正确性

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数学任务效果对比

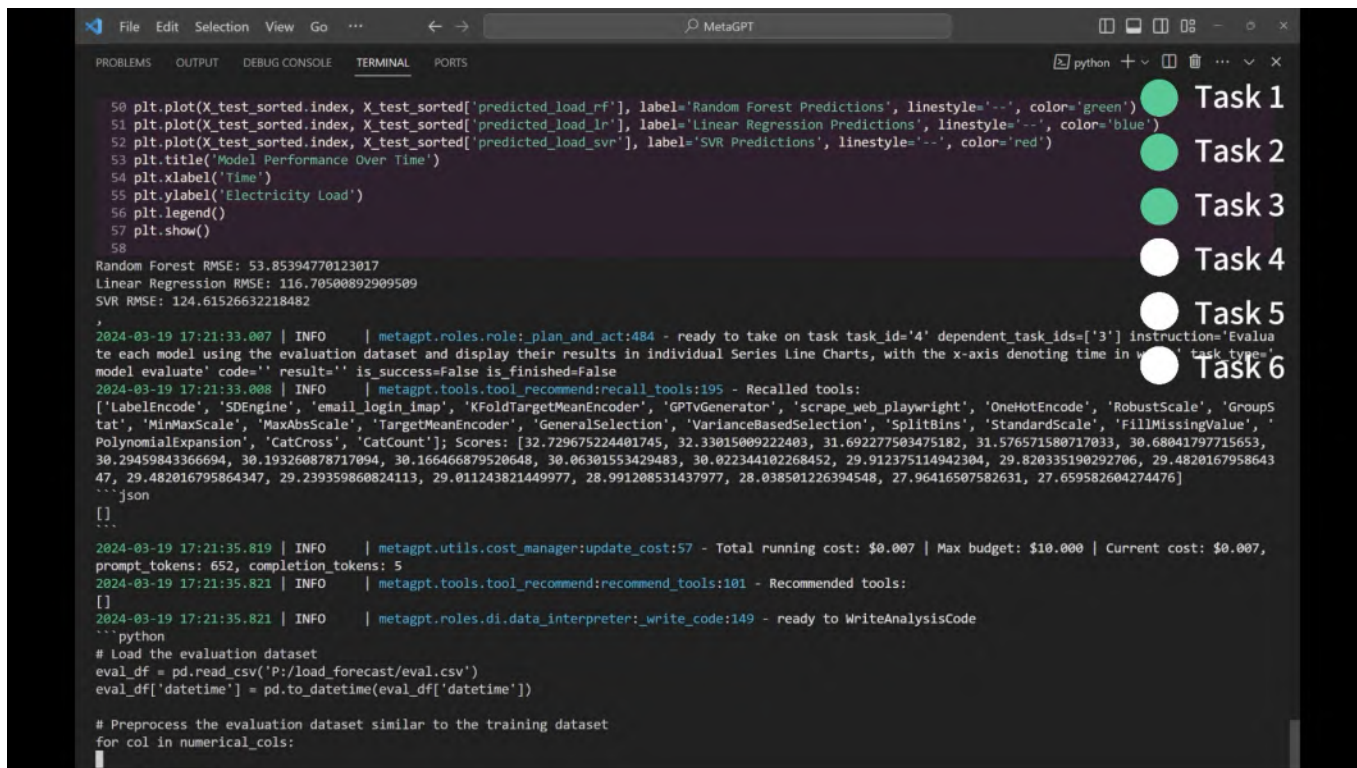


# 代码生成不可控 — 经验池 & 自动Debug

❖ **问题：** LLMs在解决复杂数据科学任务时，缺乏利用历史经验提高效率的能力

❖ **分析：** 未经优化的经验复用可能导致重复调试，降低任务执行效率

❖ **解法：** Data Interpreter构建“经验池”，存储过往任务关键元素和成功/失败尝试，通过检索相似任务经验，减少调试次数，降低成本，有效提升LLM在代码中心问题解决中的敏感度和效率



```
50 plt.plot(X_test_sorted.index, X_test_sorted['predicted_load_rf'], label='Random Forest Predictions', linestyle='--', color='green')
51 plt.plot(X_test_sorted.index, X_test_sorted['predicted_load_lr'], label='Linear Regression Predictions', linestyle='--', color='blue')
52 plt.plot(X_test_sorted.index, X_test_sorted['predicted_load_svr'], label='SVR Predictions', linestyle='--', color='red')
53 plt.title('Model Performance Over Time')
54 plt.xlabel('Time')
55 plt.ylabel('Electricity Load')
56 plt.legend()
57 plt.show()
58

Random Forest RMSE: 53.85394770123017
Linear Regression RMSE: 116.70500892909509
SVR RMSE: 124.61526632218482

2024-03-19 17:21:33.007 | INFO | metagpt.roles.role: plan_and_act:484 - ready to take on task task_id='4' dependent_task_ids=['3'] instruction='Evaluate each model using the evaluation dataset and display their results in individual Series Line Charts, with the x-axis denoting time in weeks' task_type='model evaluate' code='' result='' is_success=False is_finished=False
2024-03-19 17:21:33.008 | INFO | metagpt.tools.tool_recommender: recall_tools:195 - Recalled tools:
[{'LabelEncode', 'SDEngine', 'email_login_imap', 'KFoldTargetMeanEncoder', 'GPTvGenerator', 'scrape_web_playwright', 'OneHotEncode', 'RobustScale', 'GroupStat', 'MinMaxScale', 'MaxAbsScale', 'TargetMeanEncoder', 'GeneralSelection', 'VarianceBasedSelection', 'SplitBins', 'StandardScale', 'FillMissingValue', 'PolynomialExpansion', 'CatCross', 'CatCount'}]; Scores: [32.729675224401745, 32.33015009222403, 31.692277503475182, 31.576571580717033, 30.68041797715653, 30.29459843366694, 30.193260878717094, 30.166466879520648, 30.06301553429483, 30.022344102268452, 29.912375114942304, 29.820335190292706, 29.482016795864347, 29.482016795864347, 29.239359860824113, 29.011243821449977, 28.991208531437977, 28.038501226394548, 27.96416507582631, 27.659582604274476]
...
2024-03-19 17:21:35.819 | INFO | metagpt.utils.cost_manager: update_cost:57 - Total running cost: $0.007 | Max budget: $10.000 | Current cost: $0.007, prompt_tokens: 652, completion_tokens: 5
2024-03-19 17:21:35.821 | INFO | metagpt.tools.tool_recommender: recommend_tools:101 - Recommended tools:
...
2024-03-19 17:21:35.821 | INFO | metagpt.roles.di.data_interpreter: write_code:149 - ready to WriteAnalysisCode
...python
load the evaluation dataset
eval_df = pd.read_csv('P:/load_forecast/eval.csv')
eval_df['datetime'] = pd.to_datetime(eval_df['datetime'])

Preprocess the evaluation dataset similar to the training dataset
for col in numerical_cols:
```

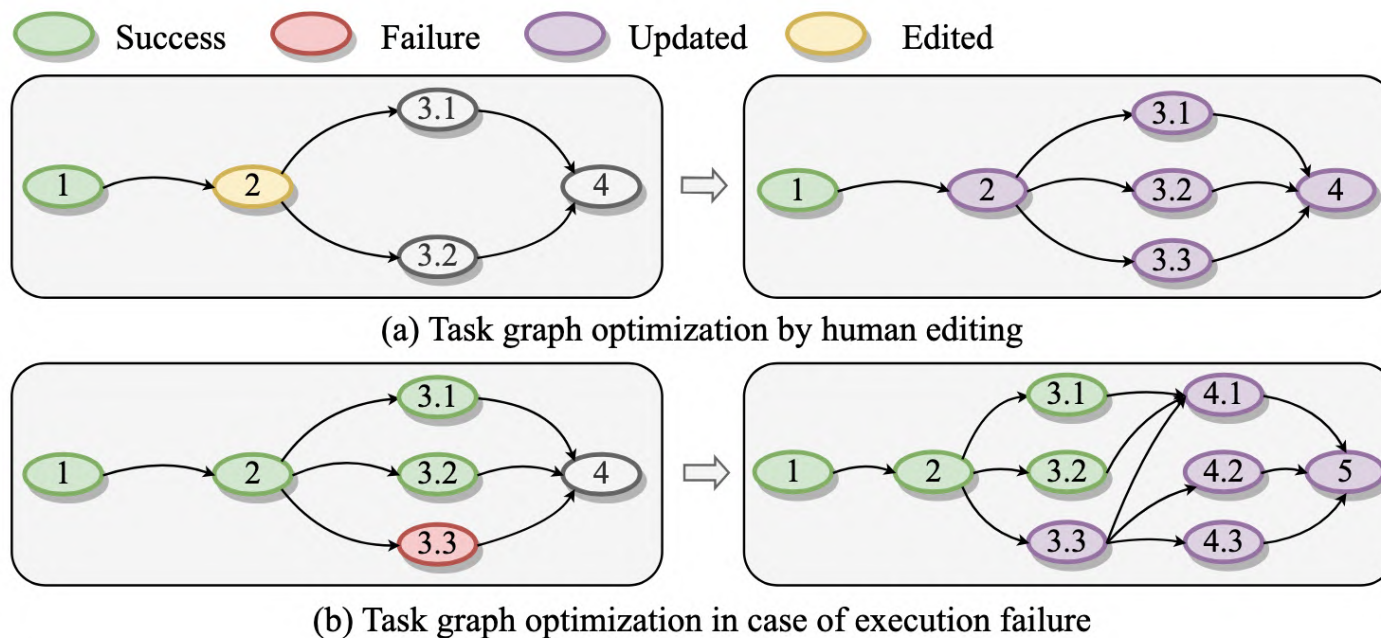
Task 1  
Task 2  
Task 3  
Task 4  
Task 5  
Task 6

# 代码生成不可控 — 经验池 & 自动Debug

❖ **问题：** LLMs在解决复杂数据科学任务时，缺乏利用历史经验提高效率的能力

❖ **分析：** 未经优化的经验复用可能导致重复调试，降低任务执行效率

❖ **解法：** Data Interpreter构建“经验池”，存储过往任务关键元素和成功/失败尝试，通过检索相似任务经验，减少调试次数，降低成本，有效提升LLM在代码中心问题解决中的敏感度和效率



# 代码生成不可控 — 经验池 & 自动Debug

❖ **问题：** LLMs在解决复杂数据科学任务时，缺乏利用历史经验提高效率的能力

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Experience pool size Metric	size=0		size=80		size=200	
	NDA	Cost (\$)	NDA	Cost (\$)	NDA	Cost (\$)
WSC	1.50	0.69	0.80	0.33	<b>0.40</b>	<b>0.29</b>
ER	1.50	0.81	0.25	0.27	<b>0</b>	<b>0.13</b>
Titanic	1.40	0.65	0.40	0.35	<b>0.40</b>	<b>0.31</b>
House Prices	1.60	1.01	0.60	0.15	<b>0.40</b>	<b>0.15</b>
ICR	1.40	0.85	0.60	0.44	<b>0.40</b>	<b>0.37</b>
Avg. / Std.	1.48 / 0.08	0.80 / 0.14	0.53 / 0.21	0.31 / 0.11	<b>0.32 / 0.18</b>	<b>0.24 / 0.10</b>

随着经验池从1增加至200，平均的debug次数从1.48降低到了0.32，执行成本从0.80美元降低到了0.24美元，这说明经验的累计对于从自然语言描述任务到代码生成能够有明显的帮助



## 更多实验结果

Table 1: **Performance comparisons on ML-Benchmark.** This table reports the comprehensive score of each task. “WR”, “BCW”, “ICR”, “SCTP”, and “SVPC” represent “Wine recognition”, “Breast cancer wisconsin”, “ICR - Identifying age-related conditions”, “Santander customer transaction prediction”, and “Santander value prediction challenge”, respectively.

Model / Task	WR	BCW	Titanic	House Prices	SCTP	ICR	SVPC	Avg.
AutoGen	0.96	<b>0.99</b>	0.87	0.86	0.83	0.77	0.73	0.86
OpenInterpreter	<b>1.00</b>	0.93	0.86	0.87	0.68	0.58	0.44	0.77
TaskWeaver	<b>1.00</b>	0.98	0.63	0.68	0.34	0.74	0.48	0.69
XAgent	<b>1.00</b>	0.97	0.42	0.42	0	0.34	0.01	0.45
<b>Data Interpreter</b>	0.98	<b>0.99</b>	<b>0.91</b>	<b>0.96</b>	<b>0.94</b>	<b>0.96</b>	<b>0.89</b>	<b>0.95</b>

Table 2: **Performance comparisons on Open-ended task benchmark.** This table reports the completion rate of each task. The tested tasks include “OCR” (Optical Character Recognition), “WSC” (Web Search and Crawling), and “ER” (Email Reply), “WPI” (Web Page Imitation), “IBR” (Image Background Removal), “T2I” (Text-to-Image), “I2C” (Image-to-Code) and “MGG” (Mini Game Generation).

Model / Task	OCR	WSC	ER	WPI	IBR	T2I	I2C	MGG	Avg.
AutoGen	0.67	0.65	0.10	0.26	<b>1.00</b>	0.10	0.20	0.67	0.46
OpenInterpreter	0.50	0.30	0.10	0.36	<b>1.00</b>	0.50	0.25	0.20	0.40
<b>Data Interpreter</b>	<b>0.85</b>	<b>0.96</b>	<b>0.98</b>	<b>1.00</b>	<b>1.00</b>	<b>1.00</b>	<b>1.00</b>	<b>0.93</b>	<b>0.97</b>

# 小结

## HoneyBee and HoneyComb

自动合成大规模高质量数据集，知识库，以及工具集可以显著提高LLM在Science领域的应用效果



## Data Interpreter

动态层次规划，整合多种工具并自动化生成和迭代工具代码集，自动验证代码，学习任务经验，这些是Data Interpreter效果提升的关键



# 极客邦科技 2024 年会议规划

促进软件开发及相关领域知识与创新的传播



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北京

全球软件开发大会

会议时间：已结束

**ArchSummit**

深圳

全球架构师峰会

会议时间：已结束

**AiCon**

上海

全球人工智能开发与应用大会

会议时间：已结束

**ArchSummit**

北京

全球架构师峰会

会议时间：12月20-21日

4月

5月

6月

8月

8月

10月

12月

**AiCon**

北京

全球人工智能开发与应用大会

会议时间：已结束

**FCon**

上海

全球金融科技大会

会议时间：已结束

**QCon**

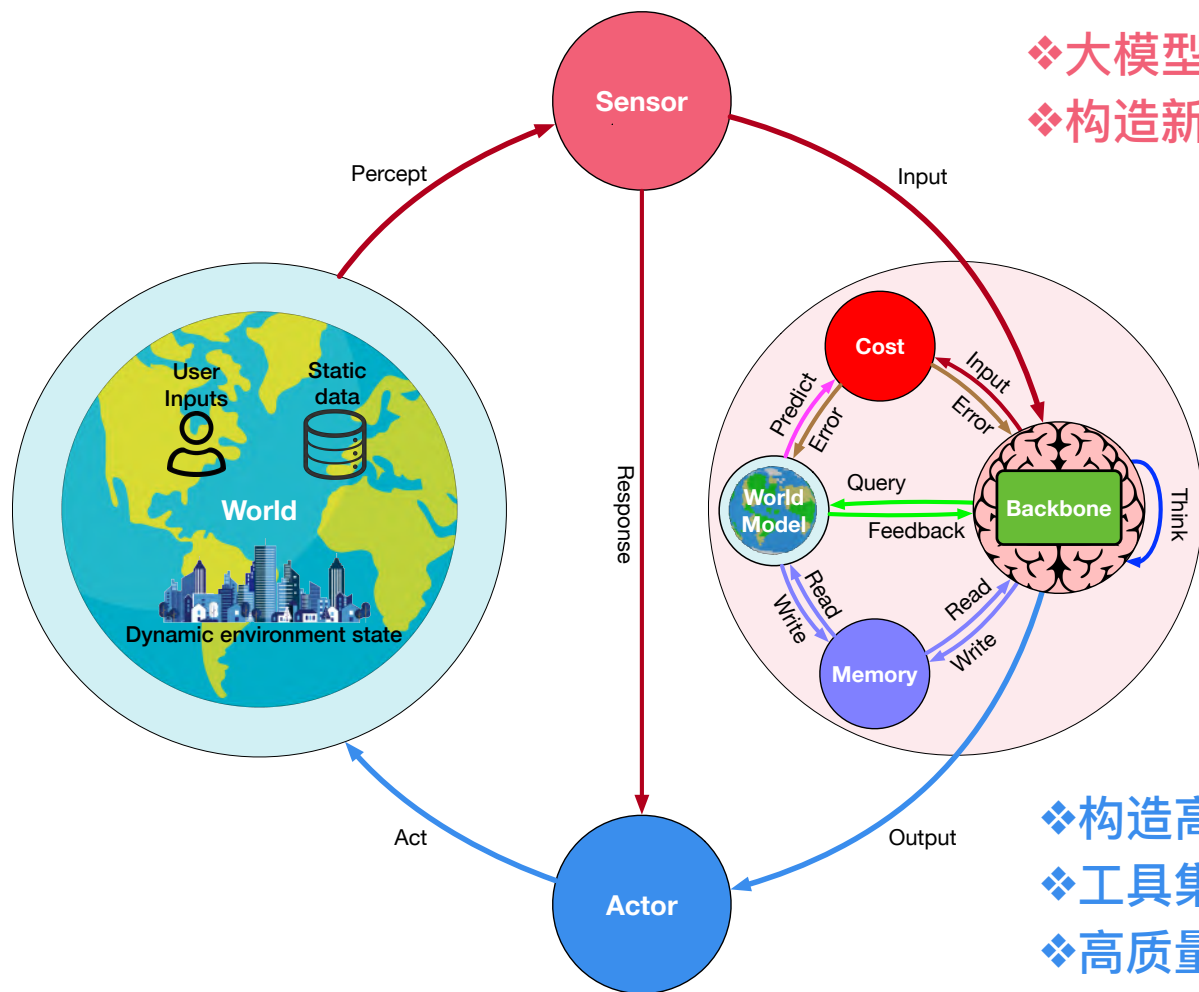
上海

全球软件开发大会

会议时间：10月18-19日



# 总结



- ❖大模型具身智能体的瓶颈在于多模态感知和动作的执行
- ❖构造新型多模态任务（e.g. VCR）有助于提高多模态表示学习

- ❖对问题的多重思考检验有助于提高推理准确性和减少幻觉
- ❖剧本杀任务可以成为一个很好的推理评测任务
- ❖通过自动构造大量高质量数据集有助于提高领域内专业知识的获取
- ❖通过化繁为简，层次图管理，复用经验的方式可以提高复杂任务的成功率

- ❖构造高质量的工具集能大大拓展智能体的能力
- ❖工具集应动态构造和优化累积
- ❖高质量的经验池也能明显提高解决复杂任务的能力

# THANKS

演讲人：刘邦

蒙特利尔大学 / Mila研究所 助理教授

Canada CIFAR AI Chair



联系方式



个人主页



B站主页